

Bayesian Inference of Hidden States and Velocities in Stochastic Spatial Movement

Abstract

Cells transport material along internal filaments in bursts of directed motion that reverse at random times, a process commonly described by a velocity-jump model in which a particle travels at one of a small number of fixed velocities and switches between them at random. Fitting such a model is difficult because a microscope records only positions, at a fixed frame rate and with measurement error, so a switch occurring between two frames leaves no direct trace. Ceccarelli, Browning and Baker (2025) address this by assuming at most m switches occur between consecutive frames, giving a group of approximations. This dissertation derives, for the case of two velocities, the exact expression that this group approaches as m grows without limit, as the distance travelled between two frames depends on the unrecorded path only through the total time spent at each velocity, so the sum over switch counts reduces to a single integral over that total time, which has a closed form in modified Bessel functions. The exact expression and the approximations at $m = 0, 1$ and 2 were implemented in R and fitted by Bayesian methods to simulated tracks with known parameters. The approximation that ignores unrecorded switches is not just imprecise but misspecified, as its likelihood is maximised 251 log units above the value at the parameters that generated the data, and its errors are undiminished on a track eight times longer. The one-switch approximation overestimates the measurement error by around 19% once switching accelerates, where the two-switch and exact treatments reach below 6%. Residual bias at the shorter track length is shown to reflect the realised trajectory rather than the method and disappears on the longer track. Where switching becomes fast relative to the frame rate, the exact likelihood remains correctly centred but nearly flat, a limit imposed by the data rather than the method.

1 Introduction

1.1 Motivation

Many biological systems move in a way that alternates between a small number of distinct modes. Bacteria alternate between running and tumbling; molecular motors alternate between processive transport along a filament and detachment; and in the developing *Drosophila* embryo, messenger RNA particles are carried along microtubules in bursts of directed motion punctuated by pauses and reversals (Ciocanel *et al.*, 2017; Harrison *et al.*, 2019). It is this last application that motivates the model studied here, and which Ceccarelli, Browning and Baker (2025) take as their example.

The quantities of biological interest are the speeds of the directed phases and the rates at which the particle switches between them, since together these determine how quickly material is delivered and how tightly it is localised. Neither can be measured directly. What is measured is a sequence of particle positions, recorded at a fixed frame rate and subject to localisation error from the imaging process. The task is to recover the parameters of the motion from this noisy, discretely sampled record.

At any instant the particle is travelling at one of its possible velocities, but the microscope does not report which. It reports position only. The velocity in force at a given moment does influence what is recorded, since it determines how far the particle travels before the next frame, but it is never itself measured and must be reconstructed from its cumulative effect on position. A quantity of this kind, which is not observed but which governs what is observed, is called a hidden state, and a model in which a hidden state evolves as a Markov process (a process whose future evolution depends only on its present state and not on the path by which it arrived there, as set out in Section 2.1) while generating observable quantities is called a hidden Markov model. Throughout this report, the hidden state is the velocity in force at a given time.

Three features of the data compound the difficulty. The hidden state is never observed directly. Observations arrive at discrete times, so a switch occurring between two frames leaves no direct trace; the particle may reverse and reverse back before the next frame, and the record will show only a displacement. And measurement error means a displacement consistent with one velocity might equally have arisen from a different velocity plus noise. Any likelihood for such data must therefore account for events that were not observed and, in principle, cannot be.

1.2 Related work

A simpler problem with a complete solution

It is helpful to begin with a closely related problem for which a full treatment already exists. Ion channels are protein pores in a cell membrane that open and close spontaneously. When open, a channel admits ions and so passes a small electrical current that can be measured directly; when closed, it passes none. A recording from a single channel is therefore close to a step function alternating between two current levels, with the transitions between them occurring at random times governed by a Markov process.

Siekmann *et al.* (2011) give a complete Markov chain Monte Carlo treatment of this problem. Their method handles the continuous-time structure of the underlying process exactly, as the probability of moving between hidden states across an observation interval is obtained from a matrix that summarises the switching rates, the generator, in a way that accounts for every possible number of intermediate transitions. Section 2.1 gives the construction. Earlier work on the statistical analysis of such recordings includes Colquhoun and Hawkes (1982) and Ball and Rice (1992).

What makes this problem tractable is the relationship between the hidden state and the observation. The recorded current level is the hidden state, corrupted by noise. Once the state at a given observation time is known, the distribution of the recorded value follows immediately, as it is a normal distribution centred on the corresponding current level. The probability of an observation given the state is called the emission density, and in this setting, it is a simple Gaussian about a fixed level.

Velocity-jump models: the additional difficulty

The problem studied here differs in one structural respect that changes the analysis substantially. In the ion-channel setting the observed quantity is piecewise constant, and each observation corresponds to a single state. In a velocity-jump model the observed quantity is position, and it is the

velocity that is piecewise constant. Position is the accumulated integral of velocity, so the hidden state determines the rate at which the observation changes, not the observation itself.

The consequence is that a single observation does not correspond to a single hidden state. Consider an interval between two frames during which the particle switches from the forward state to the backward state partway through. It travels forward for part of the interval and backward for the rest, and the position recorded at the end of the interval reflects both phases. The displacement over that interval therefore depends not only on which velocity is in force at the end, but on how many switches occurred within the interval and when they occurred. Such a displacement can take any value between the two single-velocity displacements, and no single velocity accounts for it. The emission density is consequently not a simple Gaussian about a fixed value, and constructing it is the central technical problem.

Velocity-jump models have been used across a wide range of applications, including bacterial chemotaxis (Berg and Brown, 1972; Erban and Othmer, 2004), the motion of molecular motors (Hughes *et al.*, 2011, 2012), axonal transport (Bressloff and Newby, 2013; Xue and Jameson, 2017) and RNA motility (Ciocanel *et al.*, 2017; Harrison *et al.*, 2019). The two-state case, in which a particle moves forward or backward, was treated by Rosser *et al.* (2013) and Harrison and Baker (2018), and it is their results that Ceccarelli *et al.* (2025) generalise to networks of n states.

Ceccarelli *et al.* (2025) address the emission problem by integrating over the unknown switch times. Because the resulting sum over all possible numbers of within-interval switches has no closed form in general, they truncate it at a chosen order m , giving expressions that are exact for intervals containing up to m switches and approximate beyond that. They report that the resulting estimates are accurate across a range of switching rates but degrade when switching becomes fast relative to the observation interval, and they attribute the residual uncertainty in their estimates to the limited quantity of data available. A subsequent paper (Ceccarelli *et al.*, 2026) develops a full Bayesian framework on this basis, extends it to model selection, and examines the conditions under which the parameters of such models can be recovered from data at all.

An alternative strategy avoids constructing a likelihood at all. Likelihood-free methods, including particle filtering and pseudo-marginal Markov chain Monte Carlo (Andrieu *et al.*, 2010; Warne *et al.*, 2020; Simpson *et al.*, 2022), estimate the likelihood by simulation rather than evaluating it. These methods are general but computationally demanding, and an analytically tractable likelihood, where one can be obtained, is preferable.

The gap addressed here

The two bodies of work make opposite approximations. The recursion used for ion channels, which carries a probability distribution over the hidden state forward from one observation to the next and is set out in Section 2.3, is exact about how many state changes occur across an interval but applied directly to a velocity-jump process, is wrong about what displacement they produce, because it retains an emission density appropriate to a piecewise-constant observation. The truncated hierarchy is correct about the displacement produced by a given number of switches but caps that number. Neither is exact in both respects.

1.3 Aim and objectives

The aim is to derive and evaluate a likelihood for the two-state velocity-jump model that is exact in both the transition structure and the emission, and to establish the conditions under which the existing approximations are adequate. The objectives are:

- To implement the truncated hierarchy of Ceccarelli *et al.* (2025) at orders $m = 0, 1$ and 2 , as reference approximations.
- To derive the limit of that hierarchy as m grows without bound, in closed form for the two-state case, and to confirm the derived expression by comparing it against quantities computed by independent routes that do not use the derived formula, such as direct simulation of the underlying process.
- To embed all four likelihoods in a common Bayesian sampler, so that any difference in the resulting estimates is attributable to the likelihood alone.
- To characterise, through simulation across switching regimes and track lengths, when the approximations succeed and when they fail, and to distinguish failures of the method from limits imposed by the data.

1.4 Structure of this report

Section 2 develops the methods: the Markov model, the observation model, the four likelihoods, the derivation of the exact one, and the Bayesian and computational machinery used to fit them. Section 3 describes the simulated datasets. Section 4 reports the results, Section 5 interprets them, and Section 6 states the conclusions, followed by a self-evaluation of the project's conduct and the Appendix which collects supporting diagnostic plots.

2 Methods

2.1 Continuous-time Markov chains

A continuous-time Markov chain (CTMC) occupies one of a finite set of states and moves between them at random times, with the defining property that its future evolution depends only on its current state and not on how it arrived there. The remainder of this report is restricted to the two-state case, in which the particle moves either forward (state 1) or backward (state 2). This is the setting for which the exact expression of Section 2.5 can be obtained in closed form, and it is the case treated by Rosser *et al.* (2013) and Harrison and Baker (2018) and used as the worked example by Ceccarelli *et al.* (2025).

Write $S(t)$ for the state at time t , taking the value 1 or 2. The defining property above has a concrete consequence for a two-state chain, as once the state is known, the time until the next switch has the same distribution regardless of how long the chain has already been in that state. A chain that has just entered state 1 and a chain that has occupied it for some time face the same distribution of remaining waiting time.

This memorylessness forces the waiting times to be exponential. Let λ_i be the rate at which the chain leaves state i . Then

$$P(\text{time in state } i > t) = e^{-\lambda_i t}, \quad \mathbb{E}[\text{time in state } i] = \frac{1}{\lambda_i}$$

so a larger rate means a shorter expected stay. The chain is summarised by its generator matrix

$$Q = \begin{pmatrix} -\lambda_1 & \lambda_1 \\ \lambda_2 & -\lambda_2 \end{pmatrix}$$

whose off-diagonal entries are the rates of moving between states and whose diagonal is set so that each row sums to zero. With only two states every switch necessarily goes to the other state, so no separate matrix of switch destinations is required.

Two consequences are used later. First, the probability of occupying each state after a time Δt , starting from each state, is given by the matrix exponential

$$A(\Delta t) = \exp(Q \Delta t), \quad A_{ij}(\Delta t) = P(S(t + \Delta t) = j \mid S(t) = i)$$

so that the entry in row i and column j is the probability of being in state j after Δt given that the chain was in state i at the start. It is computed here by eigendecomposition of Q .

The matrix exponential accounts for every possible number of intermediate switches, without approximation. Second, the chain has a stationary distribution π , the long-run proportion of time spent in each state, which satisfies $\pi Q = 0$ and for two states is

$$\pi = \frac{(\lambda_2, \lambda_1)}{\lambda_1 + \lambda_2}$$

The chain spends more time in the state it leaves more slowly, so the stationary distribution depends on the rates in inverse proportion. Note that it depends on the rates only through their ratio, as the pairs $(\lambda_1, \lambda_2) = (1, 0.5)$ and $(10, 5)$ give the same stationary distribution. This observation is returned to in Section 5.2.

2.2 The velocity-jump process and the observation model

The particle moves in one dimension, at velocity v_1 when the chain is in state 1 and v_2 when it is in state 2. Position is the accumulated integral of velocity,

$$x(t) = x(0) + \int_0^t v_{S(\tau)} d\tau$$

Because the velocity is piecewise constant, this integral is a sum of velocity multiplied by duration over the intervals between switches. In particular, if the chain occupies state 1 for a total time T_1 during an interval of length Δt and state 2 for the remainder, the displacement over that interval is

$$\Delta x = v_1 T_1 + v_2 (\Delta t - T_1)$$

regardless of how many switches occurred or when. This identity is the basis of the exact likelihood derived in Section 2.5.

Observations and increments

Positions are recorded on a regular grid $t_i = i\Delta t$ for $i = 0, 1, \dots, N$, each subject to independent normal measurement error:

$$y_i = x(t_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

with ε_i independent of ε_j for $i \neq j$. Inference is carried out not on the positions but on the observed increments,

$$\Delta y_i = y_i - y_{i-1}, \quad i = 1, \dots, N$$

Substituting the observation equation gives

$$\Delta y_i = [x(t_i) + \varepsilon_i] - [x(t_{i-1}) + \varepsilon_{i-1}] = \Delta x_i + (\varepsilon_i - \varepsilon_{i-1})$$

where Δx_i is the true displacement. The noise attached to an increment is therefore a difference of two independent errors, and its variance follows from the standard result for a linear combination of random variables:

$$\text{Var}(\varepsilon_i - \varepsilon_{i-1}) = \text{Var}(\varepsilon_i) + \text{Var}(\varepsilon_{i-1}) - 2 \text{Cov}(\varepsilon_i, \varepsilon_{i-1}) = \sigma^2 + \sigma^2 - 0 = 2\sigma^2$$

the covariance vanishing because the errors are independent. The emission standard deviation used throughout is therefore $\sqrt{2}\sigma$, not σ . Working with increments removes the unknown starting position from the problem and makes the observed series stationary.

2.3 The likelihood of an observed track

The hidden state is never observed, so the probability of the observed data must be computed with the states averaged out. This is the standard hidden Markov model problem, solved by a forward recursion (Rabiner, 1989).

The components of a hidden Markov model

The model now has two layers. The first is the hidden layer, the sequence of states occupied at each observation time, which evolves according to the Markov chain of Section 2.1 and is never seen. The second is the observed layer, the sequence of increments, each generated from the hidden state that produced it. A hidden Markov model is specified by three things: the initial distribution over the hidden state at the first observation, taken here to be the stationary distribution; the transition structure, which governs how the hidden state evolves between observations and is given by the generator of Section 2.1; and the emission density, which gives the probability of an observation for each hidden state.

The four likelihoods compared in this report share the first two and differ only in the third. The difficulty specific to the velocity-jump setting, as Section 1.2 described, is that the emission cannot be a simple density attached to a single state, since a switch within an interval means the observation reflects two states rather than one. The emission is therefore written $f_i(s \rightarrow s')$, indexed by both the state at the start of the interval and the state at its end.

Let $\alpha_i(s)$ be the probability that the chain is in state s at the start of interval i , given the increments observed so far, and let $f_i(s \rightarrow s')$ be the emission density: the probability density of observing increment Δy_i given that the interval began in state s and ended in state s' . The recursion is

$$\tilde{\alpha}_i(s') = \sum_s \alpha_{i-1}(s) f_i(s \rightarrow s'), \quad c_i = \sum_{s'} \tilde{\alpha}_i(s'), \quad \alpha_i(s') = \frac{\tilde{\alpha}_i(s')}{c_i}$$

initialised at the stationary distribution, and the log-likelihood is the accumulated log of the scaling constants,

$$\log L(\theta) = \sum_i \log c_i$$

Rescaling at each step is necessary for numerical stability, as without it the forward variable is a product of N densities and underflows to zero within a few hundred intervals. The essential feature is that the hidden states are marginalised rather than estimated. At no point does the algorithm decide which state the particle occupied; it carries a probability distribution over both possibilities and lets each observation reweight it.

All four likelihoods share this recursion and differ only in the emission density $f_i(s \rightarrow s')$, that is, in how they treat what may have happened within an interval.

2.4 Three approximate emissions

A switch occurring between two observations has two consequences: it changes which state holds at the next observation time, and it changes the displacement recorded over the interval. The three likelihoods of Ceccarelli *et al.* (2025) implemented here treat these to differing degrees. Table 1 summarises the resulting group.

Option A: no within-interval switching

Option A assumes no switch occurs inside an observation interval, so each interval consists of motion at a single velocity and the emission is

$$f(\Delta y | s) = N(\Delta y; v_s \Delta t, 2\sigma^2)$$

Transitions between intervals are handled exactly through $A(\Delta t) = \exp(Q\Delta t)$, so the recursion takes the form $\alpha_i = [\alpha_{i-1} A(\Delta t)] \odot f_i$, with $\odot$ denoting elementwise multiplication. Option A is therefore exact in its treatment of the switch count and approximate in its treatment of the displacement. It corresponds to applying the ion-channel machinery of Section 1.2 directly to a velocity-jump process.

Option B: at most one within-interval switch

If exactly one switch occurs, from state s to state s' at time τ after the start of the interval, the displacement is

$$\Delta x(\tau) = v_s \tau + v_{s'}(\Delta t - \tau), \quad \tau \in (0, \Delta t)$$

which is linear in τ and therefore confined to the range between $v_1 \Delta t$ and $v_2 \Delta t$. The density of τ given exactly one switch is proportional to $\lambda_{-s} e^{-\lambda_{-s} \tau} e^{-\lambda_{-s'}(\Delta t - \tau)}$, the chain leaving s at time τ and then remaining in s' for the rest of the interval. Its normalising constant has the closed form

$$Z_1 = \frac{\lambda_1 e^{-\lambda_2 \Delta t} (1 - e^{-(\lambda_1 + \lambda_2) \Delta t})}{\lambda_1 - \lambda_2} \quad (\lambda_1 \neq \lambda_2), \quad Z_1 = \lambda_1 e^{-\lambda_1 \Delta t} \Delta t \quad (\lambda_1 = \lambda_2)$$

Changing variables from τ to Δx and convolving with the measurement noise gives the one-switch emission. The recursion weights the zero-switch and one-switch terms by their probabilities.

Option C: at most two within-interval switches

Two switches in a two-state chain return the particle to its starting state, so the displacement depends only on w , the total time spent in the other state, through $\Delta x = v_- s \Delta t + (v_- \{s'\} - v_- s)w$. Marginalising the joint density of the two switch times over their start point gives a closed-form density for w , again convolved with the noise. Following Ceccarelli *et al.* (2025, Equation 5), the three terms are weighted by the probabilities of exactly zero switches, exactly one, and two or more, so that the top term carries the whole remaining tail.

Table 1. The four emissions, by treatment of the switch count and the displacement. Options A, B and C follow Ceccarelli *et al.* (2025); Option D is derived in Section 2.5.

Option	Switch count	Displacement
A	Exact, via $\exp(Q\Delta t)$	Single velocity per interval
B	Truncated at 1	Exact for 0 or 1 switch
C	Truncated at 2	Exact for 0, 1 or 2 switches
D	Exact, all counts	Exact for all counts

2.5 Option D: an exact emission via the occupation time

As established in Section 2.2, the displacement over an interval depends on the hidden path only through the total time spent in the interval's starting state. The number of switches and their individual times do not enter. Summing over switch counts, as the truncated hierarchy does, is therefore unnecessary, as it suffices to marginalise over the distribution of that occupation time.

The derivation is presented for an interval beginning in state 1; the case of an interval beginning in state 2 follows by exchanging the indices 1 and 2 throughout. Write u for the time spent in state 1 and $w = \Delta t - u$ for the time spent in state 2. The distribution of u has two parts. The law of the occupation time is a known result, and the two subsections below reproduce its derivation by switch count in order to make the parity structure explicit, since it is that structure, rather than the density itself, that the emission construction depends on.

The no-switch atom

With probability $e^{-\lambda_1 \Delta t}$ the chain does not switch at all, giving $u = \Delta t$ exactly and displacement $v_1 \Delta t$. This is a point mass rather than a density and is handled separately.

The continuous part

For $u \in (0, \Delta t)$ the chain switches at least once. Because a two-state chain alternates, the interval ends in state 2 after an odd number of switches and back in state 1 after an even number, so the two cases can be summed separately.

For an odd number of switches, $2k + 1$ with $k \geq 0$, the chain makes $k + 1$ visits to state 1 and $k + 1$ to state 2, of which k visits to each are complete and contribute Erlang densities in u and w respectively, the Erlang density being the distribution of a sum of independent exponential waiting times. Summing the product over all k :

$$\sum_{k \geq 0} \frac{\lambda_1^{k+1} u^k}{k!} \cdot \frac{\lambda_2^k w^k}{k!} = \lambda_1 \sum_{k \geq 0} \frac{(\lambda_1 \lambda_2 u w)^k}{(k!)^2} = \lambda_1 I_0(z)$$

where $z = 2\sqrt{\lambda_1 \lambda_2 u w}$ and I_k denotes the modified Bessel function of the first kind of order k , defined by the series

$$I_k(z) = \sum_{j \geq 0} \frac{(z/2)^{2j+k}}{j! (j+k)!}$$

The infinite sum over switch counts therefore collapses to a single special function. For an even number of switches, $2j$ with $j \geq 1$, the corresponding sum is

$$\sum_{j \geq 1} \frac{\lambda_1^j u^j}{j!} \cdot \frac{\lambda_2^j w^{j-1}}{(j-1)!} = \sqrt{\frac{\lambda_1 \lambda_2 u}{w}} I_1(z)$$

Attaching the exponential factor for the total time spent in each state and combining the two cases gives the occupation-time density

$$g(u) = e^{-\lambda_1 u - \lambda_2 w} \left[\lambda_1 I_0(z) + \sqrt{\frac{\lambda_1 \lambda_2 u}{w}} I_1(z) \right], \quad u \in (0, \Delta t)$$

This is the standard occupation-time density for a two-state continuous-time Markov chain (Pedler, 1971). The Bessel functions arise as the closed-form sum of the series over switch counts, so that the j -th term of I_k corresponds to paths making a particular number of switches, as the special function is precisely the infinite sum that the truncated hierarchy approximates by stopping early. Both series identities above were confirmed numerically against explicit term-by-term summation.

The end state is retained, not marginalised away. The two Bessel terms are not just two components of a sum to be added together. The I_0 term collects the odd switch counts and therefore carries exactly the probability mass that ends in state 2; the I_1 term collects the even counts and carries the mass that ends back in state 1. Keeping them separate yields the emission density resolved by end state, $f(\Delta y \mid s \rightarrow s')$, which is precisely what the forward recursion of Section 2.3 requires. Option D marginalises over the switch count and the switch times, which are nuisance quantities carrying no information about the parameters beyond what the occupation time already carries, while preserving in full the start-state to end-state routing. It is a complete forward recursion, not a simplification of one.

Pedler (1971) gives the total occupation-time density; the decomposition by switch-count parity used here, and its interpretation as an emission resolved by end state, is what allows that density to be used within the forward recursion.

Mapping u to Δx through the identity of Section 2.2 and convolving each part with the measurement noise gives an emission tensor that routes probability mass from start state to end state directly. No

transition matrix is required, because the occupation-time density already encodes the transition structure.

2.6 Numerical evaluation

The occupation-time density is available in closed form, but its convolution with the normal measurement noise is not. Writing φ for the $N(0, 2\sigma^2)$ density, the emission is

$$f(\Delta y \mid s \rightarrow s') = \int_0^{\Delta t} g_{s'}(u) \varphi(\Delta y - \Delta x(u)) du$$

taken over $u \in [0, \Delta t]$ and evaluated by the composite Simpson rule, which approximates an integral over $[a, b]$ divided into an even number n of subintervals of width $h = (b-a)/n$ as

$$\int_a^b f(x) dx \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + \dots + 4f_{n-1} + f_n]$$

The default grid uses 513 points. The same rule is used for the one-switch and two-switch emissions of Options B and C over the compact supports of their densities at 129 points. All four likelihoods are therefore exact in derivation up to their stated switch-count truncation and numerically approximate at the final integration step; sensitivity to the grid is reported in Section 4.10.

For efficiency the emission densities are precomputed for all increments before the recursion begins, since they depend only on the observed increments and the parameters and not on the forward variable. The convolution is then a matrix product, which is substantially faster than looping over increments and gives an identical result.

2.7 Confirming the derived expression

The derivation of Section 2.5 was confirmed by comparing it against quantities computed by routes that do not use the derived formula, so that an error in the derivation could not propagate into its own check.

The principal reference is direct simulation. Four hundred thousand CTMC paths were simulated from their exponential holding times alone, using $\lambda_1 = 1$, $\lambda_2 = 0.5$, $\Delta t = 0.3$ and starting in state 1, and for each path the time spent in state 1 and the state occupied at the end of the interval were recorded. This uses no part of the Bessel derivation. Three quantities were compared against it: the probability of no switch, and the occupation-time density separately for each end state. Two further checks are internal: the two Bessel terms must sum to the total density, and the emission must integrate to one over the increment.

Comparison against the one-switch emission

A final check compares Option D against Option B, which was implemented independently. Setting $k = 0$ in the odd-switch series of Section 2.5 leaves the single term $\lambda_1 e^{-\lambda_1 u} e^{-\lambda_2(\Delta t - u)}$, which is the density corresponding to exactly one switch. Adding the no-switch atom to it gives the emission of a process restricted to at most one switch per interval, which is precisely what Option B constructs by a different route, so the two should agree.

It is worth being explicit that this uses the leading term of the odd series, which ends in the opposite state and corresponds to one switch, and not the even series, which requires at least two. Agreement is assessed as the integration grid is refined, so that any residual difference can be attributed to discretisation rather than to the derivation. Results are reported in Section 4.1.

2.8 The Bayesian model

Bayesian inference treats the unknown parameters as random variables and combines the information in the data with prior beliefs to give a posterior distribution,

$$p(\theta | y) \propto L(y | \theta) p(\theta)$$

where $L(y | \theta)$ is one of the four likelihoods and $p(\theta)$ the prior. The output is a distribution over parameter values rather than a single estimate, so uncertainty is quantified directly. The parameter vector is

$$\theta = (v_1, v_2, \log \lambda_1, \log \lambda_2, \log \sigma)$$

The rates and the noise scale enter on the logarithmic scale. Ceccarelli *et al.* (2025) adopt the same treatment for the rates, their stated reason being that switching times are exponentially distributed, so the rates are naturally multiplicative quantities; they sample the noise scale on its natural scale, bounded above by 100. Extending the logarithmic treatment to the noise scale is a departure adopted here. Working on the log scale keeps both quantities positive without a rejection step, and makes a symmetric proposal multiplicative rather than additive, which suits parameters whose plausible values span orders of magnitude.

Table 2. Prior distributions. Specified directly on the sampled scale, so that no Jacobian adjustment is required.

Parameter	Prior	Interpretation
v_1, v_2	$N(m_v, 5000^2)$	Very broad: the standard deviation exceeds the true velocities, so the prior is close to flat over any plausible range
$\log \lambda_i$	$N(0, 2^2)$	Median rate of 1, covering roughly 0.02 to 50 within two standard deviations
$\log \sigma$	$N(\log 50, 1^2)$	Median noise scale of 50, covering roughly 7 to 370

The velocity prior is centred at m_v , a location derived from the observed increments as described in Section 2.9, rather than at a fixed value; its width is such that the prior contributes negligible information about the velocities. Ceccarelli *et al.* (2025) instead use uniform priors with hard bounds, namely $[0, 4000]$ and $[-3000, 0]$ for the velocities, $[-4, 4]$ for each log rate and $[0, 100]$ for the noise scale. The normal priors used here place no hard bound on any parameter, which as Section 5.7 shows is what allows one failure mode to become visible.

The ordering constraint

Nothing in the model specifies which of the two states is called state 1. If the labels are exchanged, and the velocities and rates exchanged with them, the resulting model assigns exactly the same probability to the observed data. The likelihood is therefore invariant under relabelling, and the posterior has two identical modes that are mirror images of one another.

This is a nuisance rather than a genuine ambiguity, but it has a practical cost. When several chains are run, some may settle on one labelling and others on the reverse, and the resulting disagreement registers as a failure to converge even though every chain has found the same physical solution. This is known as label switching (Stephens, 2000; Jasra, Holmes and Stephens, 2005). It is resolved here by requiring the velocities to be ordered, assigning zero prior density whenever $v_1 \leq v_2$, which confines every chain to a single canonical labelling in which state 1 is the faster-moving state. The constraint is placed on the velocities only; once the states are fixed, the rates are not exchangeable and require no constraint.

2.9 Markov chain Monte Carlo

The posterior cannot be written down in closed form, so it is explored by simulation. Markov chain Monte Carlo (MCMC) constructs a Markov chain over parameter space whose long-run distribution is the posterior, so that the sequence of visited values can be treated as a sample from it. Because the chain begins at an arbitrary starting point, its early iterations reflect that starting point rather than the posterior; these are discarded and are referred to as the burn-in.

The Metropolis-Hastings step

From the current parameter vector ϑ , a candidate is drawn from a multivariate normal centred on it, $\vartheta^* \sim N(\vartheta, \Sigma)$. The covariance Σ is not diagonal, so proposed moves in different parameters are correlated. This is deliberate and is the purpose of the adaptive scheme described below: the posterior itself has correlated parameters, as Section 4.6 shows for the velocities and the noise scale, and a proposal that ignored this correlation would repeatedly suggest moves in directions the posterior does not extend, which would be rejected. The candidate is accepted with probability

$$\alpha = \min \left\{ 1, \frac{L(y | \theta^*) p(\theta^*)}{L(y | \theta) p(\theta)} \right\}$$

evaluated on the log scale for numerical stability. If accepted, the candidate becomes the current value; otherwise, the current value is repeated. Because the proposal distribution is symmetric, the Hastings correction for asymmetric proposals is not required.

Adaptive tuning of the proposal

The efficiency of this scheme depends on Σ . Too small, and the chain explores in steps so short that it moves slowly; too large, and almost every candidate is rejected. The appropriate scale and orientation are not known in advance, so they are learned from the chain itself, following the adaptive Metropolis algorithm of Haario, Saksman and Tamminen (2001). At intervals during burn-in, the covariance is re-estimated from the chain's recent history as

$$\Sigma = \frac{2.38^2}{d} \cdot \text{Cov}(\theta_a, \dots, \theta_t) + \epsilon I$$

where $d = 5$ is the number of parameters, the covariance is taken over the window from the start of adaptation to the current iteration, I is the identity matrix, and $\epsilon = 10^{-6}$ is a small constant that keeps the matrix numerically positive definite. The factor $2.38^2/d$ is the optimal scaling for a random-walk proposal on an approximately normal target derived by Roberts and Rosenthal (2001).

Two further controls are applied. Adaptation begins only after iteration 300, so the covariance is estimated from settled rather than transient behaviour. A lower bound is imposed on the diagonal of Σ to prevent an unrepresentative early covariance estimate from collapsing the proposal so far that the chain stops moving; the value of this bound proved consequential and is examined in Section 4.11.

Adaptation stops at the end of burn-in, so that the retained portion of each chain is a genuine Markov chain with the correct limiting distribution.

Starting values and burn-in

Several chains are run from different starting points, drawn about a common centre so that the chains begin some distance apart and their subsequent agreement is evidence of convergence rather than of a shared origin. The velocity centre is taken from the data, as the standard deviation of the observed increments divided by Δt , which gives the order of magnitude of a plausible velocity, with opposite signs for the two states; the rates and noise scale are centred at their prior medians, and the starting velocities are sorted to satisfy the ordering constraint.

2.10 Assessing convergence

Because the sampler is constructed as a Markov chain, its distribution at any given iteration depends on where the chain started and approaches the posterior only as the chain is run. The early iterations therefore reflect the starting point rather than the target, and it is necessary to establish that a chain has run long enough for this influence to have decayed.

The standard diagnostic is the Gelman-Rubin statistic (Gelman and Rubin, 1992), which compares the variability between chains started at different points with the variability within them. For m chains of length n after burn-in, with $\bar{\vartheta}_j$ the mean of chain j , $\bar{\vartheta}$ the overall mean and s_j^2 the sample variance of chain j , define the between-chain and within-chain variances

$$B = \frac{n}{m-1} \sum_j (\bar{\vartheta}_j - \bar{\vartheta})^2, \quad W = \frac{1}{m} \sum_j s_j^2$$

and from them the pooled variance estimate and the statistic

$$\widehat{\text{var}} = \frac{n-1}{n} W + \frac{1}{n} B, \quad \hat{R} = \sqrt{\frac{\widehat{\text{var}}}{W}}$$

If the chains have converged to a common distribution the two variances agree and $\hat{R}$ approaches 1; values appreciably above 1 indicate that the chains are still exploring different regions. A conventional threshold of $\hat{R} < 1.1$ is adopted here. The statistic is reported for each parameter separately rather than as a single maximum, since a run may have converged in four parameters and not the fifth, and per-chain posterior means are given wherever it is elevated.

Two qualifications on this diagnostic are needed. First, agreement between chains is evidence of convergence only if the chains are exploring the same region, as a set of chains that has settled into a single local mode may give an $\hat{R}$ close to 1 while having explored only part of the posterior. Second, an inflated $\hat{R}$ does not by itself indicate that the chains have failed to converge. If the posterior has more than one mode and different chains settle in different modes, the between-chain variance is

large by construction, and the statistic reports that difference rather than a failure to equilibrate. $\hat{R}$ is built on the premise that the chains are converging to a common distribution, so multimodality violates its premise rather than being detected by it.

For this reason, the trace plots are treated as the primary evidence and the statistic as a summary of them. A chain whose trace shows a trend has not settled. A chain that oscillates about a stable level has settled, even if it does so slowly and therefore yields correlated samples inefficiently. Where several chains agree and one does not, the traces distinguish a chain still travelling from one that has settled elsewhere.

Where the sampler locates more than one mode, the modes are reported separately rather than pooled, since a posterior mean averaged across two distinct regions describes neither. In each such case the per-chain posterior means are given, so the number of chains reaching each mode is visible, and that proportion is itself informative about the relative size of the modes.

3 Data sources

3.1 Simulation procedure

All data used in this work are simulated, so that estimates can be compared against known true values. Hidden paths are generated exactly by the Gillespie (1976) algorithm: from the current state, draw a holding time from the exponential distribution with that state's rate, switch to the other state, and repeat until the total simulated time is reached. Because the algorithm produces the switch times exactly, the hidden path is known and can be compared against what the inference recovers.

The true position is then evaluated on the observation grid by integrating the piecewise-constant velocity as in Section 2.2, independent normal noise of standard deviation σ is added at each observation time, and the resulting positions are differenced to give the increments used for inference.

3.2 Datasets

The baseline configuration matches the two-state example of Ceccarelli *et al.* (2025) in every respect, which permits direct comparison with their published results. Variant datasets modify one aspect at a time.

Table 3. Datasets used in this work. All share the velocities $v_1 = 2000$ and $v_2 = -1500$, the noise scale $\sigma = 50$, and the observation interval $\Delta t = 0.3$.

Dataset	λ	T	N	Purpose
Baseline	(1, 0.5)	60	200	Matches Ceccarelli et al. (2025)
Faster switching	(4, 2)	60	200	Strains the truncations
Fast switching	(8, 4)	60	200	Tests the limit of what the data support
Longer track	(1, 0.5)	480	1600	Eight times more observations, same frame rate
Thinned	(1, 0.5)	60	100	Baseline, every second observation retained

3.3 The baseline dataset

Figure 1 shows the simulated baseline trajectory with its hidden state sequence. The particle alternates between periods of forward and backward motion, and at this noise level the observed positions lie close to the true path.

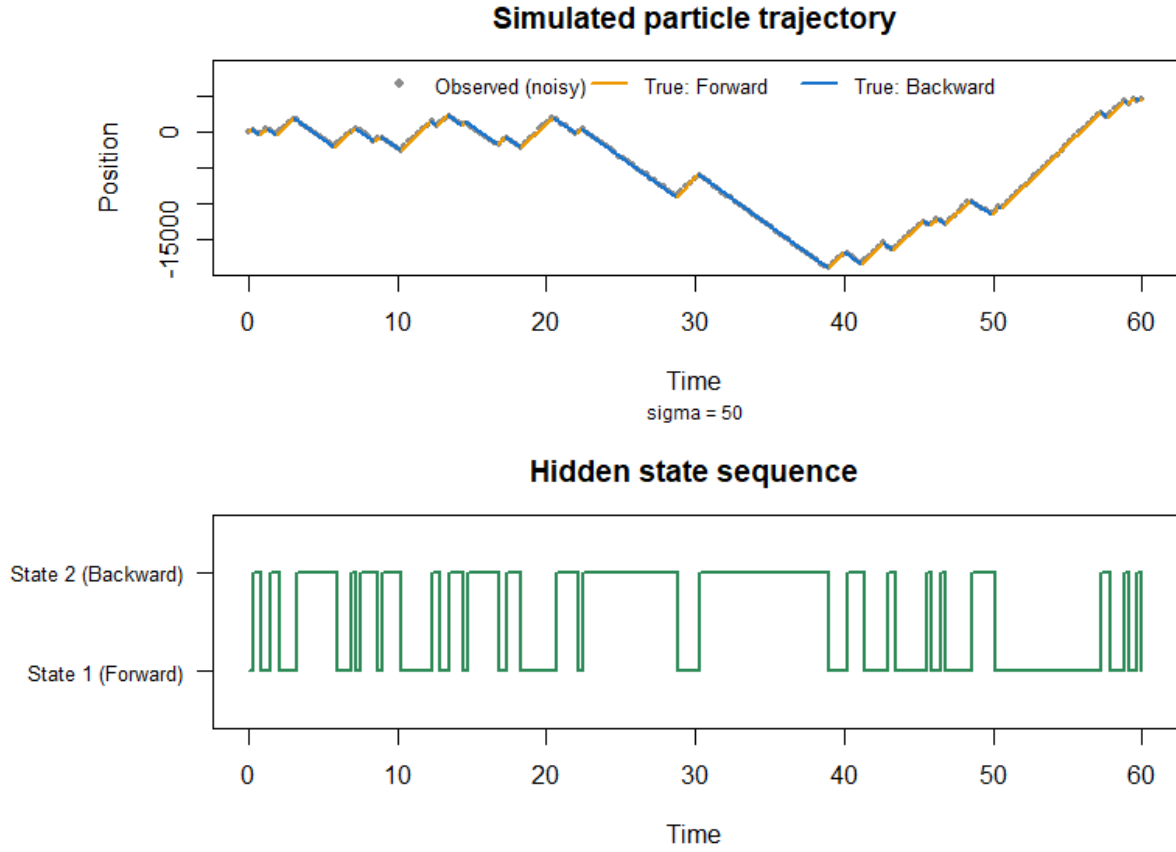


Figure 1. The simulated baseline dataset. Top: the true position coloured by hidden state, with the noisy observations in grey. Bottom: the hidden state sequence over the same window.

Figure 2 shows the distribution of the observed increments. It is bimodal, with modes at $v_1\Delta t = 600$ and $v_2\Delta t = -450$, and each component has variance $2\sigma^2 = 5000$ as derived in Section 2.2. A small number of increments fall between the two modes; these come from intervals containing at least one switch.

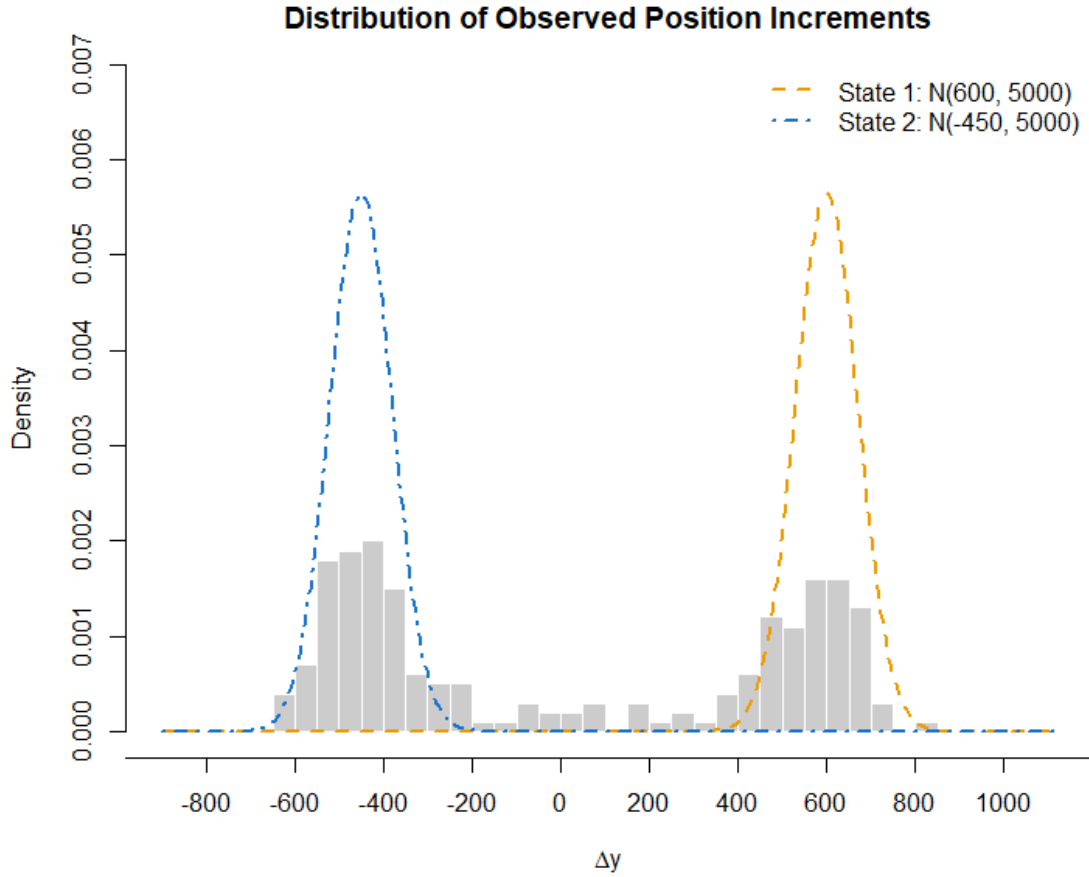


Figure 2. *Distribution of the observed position increments. The overlaid curves are the zero-switch emission densities for the two states.*

Figure 3 shows five realisations generated from the identical parameter set, differing only through the randomness of the switching times. The final positions range from -21857 to 8717 , with a standard deviation of 13424 , which is larger than most of the individual displacements.

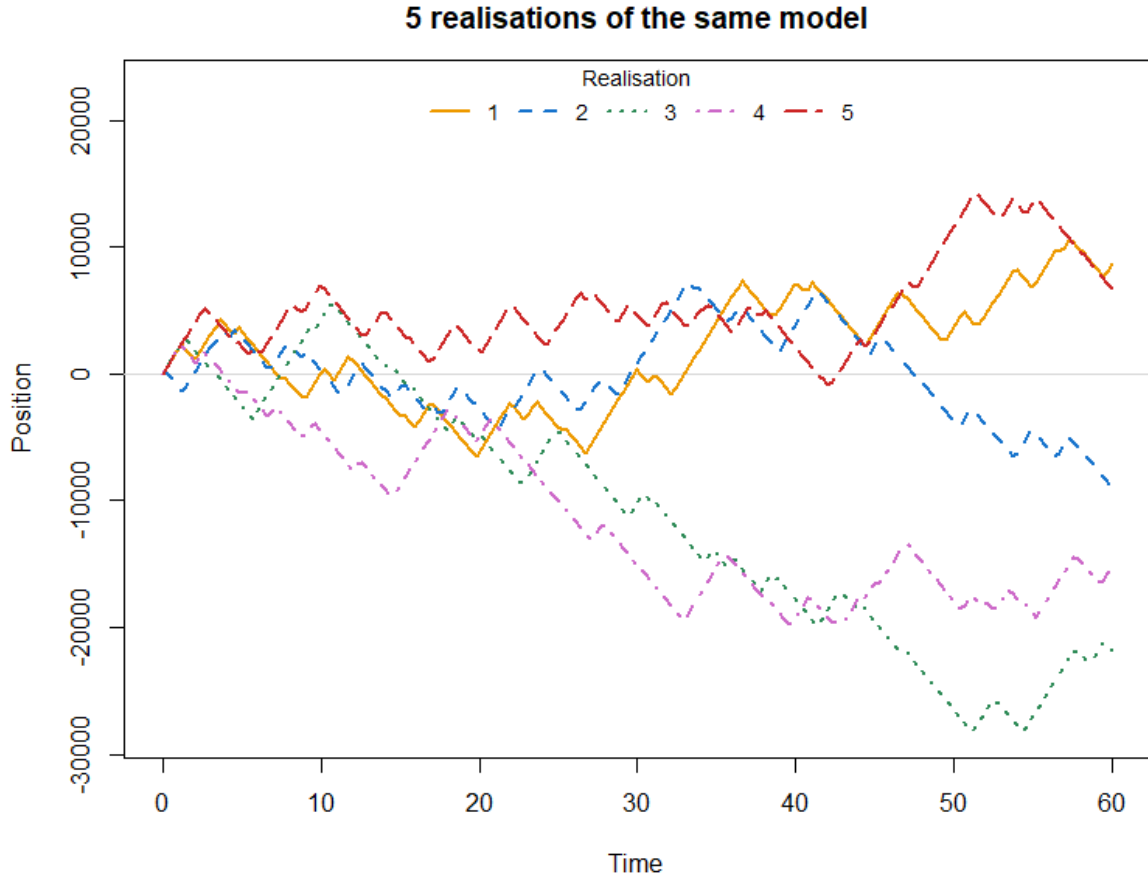


Figure 3. Five realisations of the same model with identical parameters.

3.4 Switching-event counts

Since the rates are estimated from switching events, the number of such events and their distribution across observation intervals characterises what each dataset can support. The expected number per interval is computed from the stationary distribution of Section 2.1, weighting each state's rate by the proportion of time spent in it.

Table 4. Switching-event counts by dataset. Columns give the total number of interior switches, the expected and observed switches per interval, and the number of intervals containing 0, 1, 2 or more, and 3 or more switches.

Dataset	Switches	Per interval (exp / obs)	Intervals 0 / 1 / 2+ / 3+
Baseline, $\lambda = (1, 0.5)$	50	0.20 / 0.25	154 / 42 / 4 / 0
Faster, $\lambda = (4, 2)$	168	0.80 / 0.84	87 / 72 / 41 / 12
Fast, $\lambda = (8, 4)$	309	1.60 / 1.54	48 / 60 / 92 / 40
Longer track, $N = 1600$	309	0.20 / 0.19	1320 / 254 / 26 / 3
Thinned, $k = 2$	50	0.40 / 0.50	59 / 33 / 8 / 1

In the baseline dataset the hidden path occupies state 1 at 92 of the 201 observation times and state 2 at the remaining 109, a split of 46% to 54%. The stationary distribution for these rates is 33% to 67%.

4 Results

All point values reported in this section are posterior means of the pooled post-burn-in samples across chains, and bracketed intervals are the 2.5% and 97.5% sample quantiles of the same pooled samples. A 95% credible interval is the range containing 95% of the posterior probability, so that under the fitted model the parameter lies within it with probability 0.95.

4.1 Confirming the derived expression

Before the exact likelihood is used for inference, the checks described in Section 2.7 establish that the derived expression is correct.

Table 5. Checks on the Option D emission. *The Monte-Carlo reference uses 400,000 simulated paths with $\lambda_1 = 1$, $\lambda_2 = 0.5$, $\Delta t = 0.3$, starting in state 1.*

Check	Result
Probability of no switch, analytic against simulation	0.74082 against 0.74188
Occupation density against simulation, ending in state 1	mean absolute error 0.0033
Occupation density against simulation, ending in state 2	mean absolute error 0.0131
Two Bessel terms sum to the total density	maximum error 1.11×10^{-16}
Emission integrates to 1 over the increment	0.9997 and 0.9998 for the two start states

Table 6 reports the comparison against the independently implemented one-switch emission of Option B, described in Section 2.7. The difference falls by approximately half at each doubling of the integration grid, consistent with the residual being discretisation error rather than an error in the derivation.

Table 6. One-switch truncation of Option D against the Option B emission. *Maximum relative difference over 14 test increments, as the integration grid is refined.*

Grid points	Maximum relative difference
201	0.0323
513	0.0126
1025	0.0063
4097	0.0016

4.2 Convergence at the baseline

Table 7 gives the Gelman-Rubin statistic for each parameter at the baseline dataset, establishing that the posterior summaries that follow can be relied upon. All values are below 1.05. Options A, B and D were run with four chains of 6,000 iterations. Option C on those settings gave $\hat{R} = 1.115$ for $\log \lambda_2$; it was re-run with six chains of 12,000 iterations, and the values below are from that longer run.

Table 7. Gelman-Rubin $\hat{R}$ by parameter at the baseline dataset.

Option	v_1	v_2	$\log \lambda_1$	$\log \lambda_2$	$\log \sigma$	Chains
A	1.0013	1.0004	1.0012	1.0002	1.0039	4
B	1.0005	1.0029	1.0047	1.0100	1.0025	4
C	1.0017	1.0033	1.0037	1.0088	1.0030	6
D	1.0008	1.0026	1.0073	1.0479	1.0091	4

Figure 4 shows the post-burn-in trace plots for Option B. The four chains overlap for all five parameters with no visible drift or separation. Trace plots for the other three options are given in the Appendix.

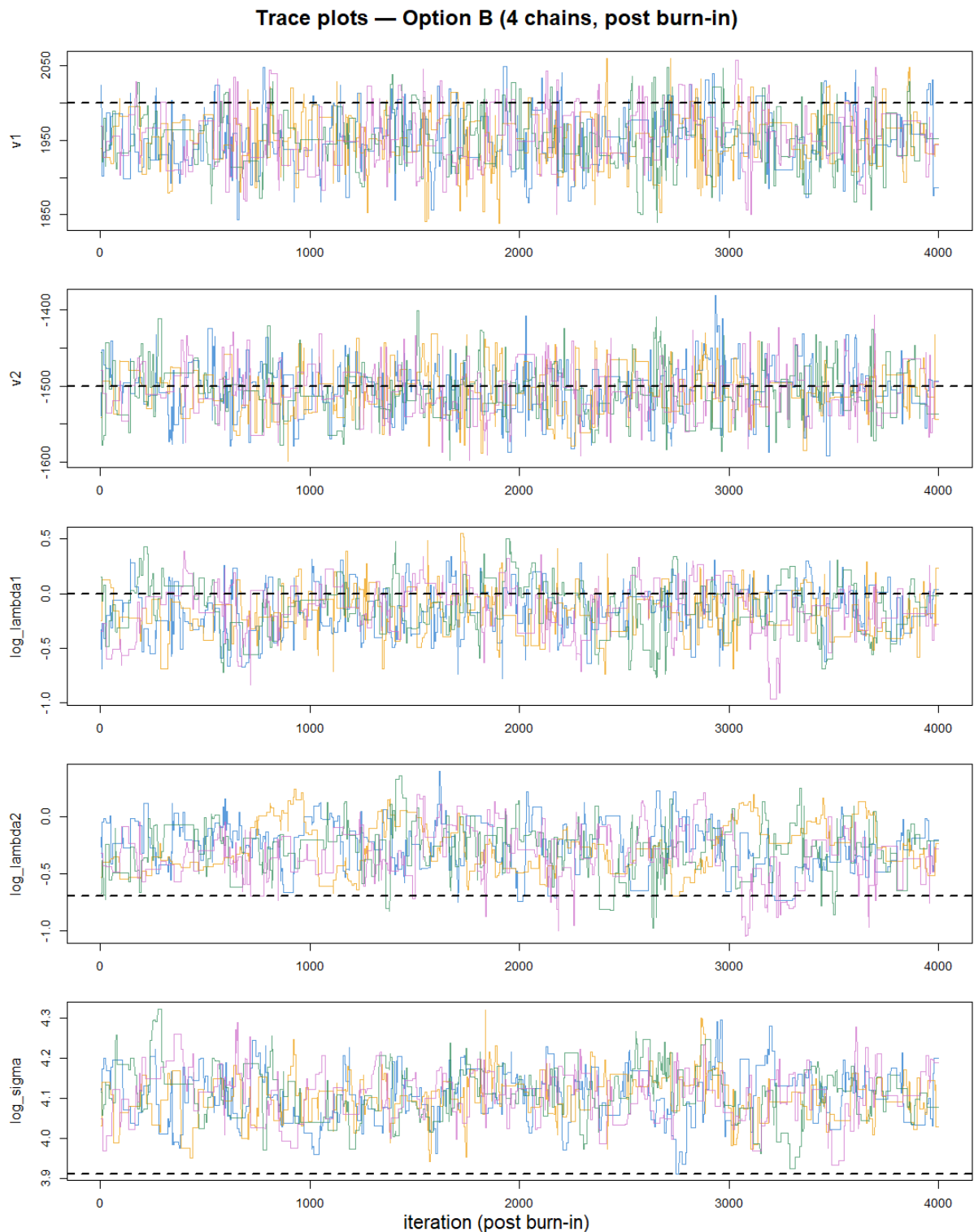


Figure 4. Trace plots for Option B, four chains, post burn-in. Dashed lines mark the true values.

4.3 Baseline parameter recovery

This section establishes how well each likelihood recovers parameters from the baseline dataset, in which switching is slow relative to the frame rate.

Table 8. Baseline recovery, $N = 200$, $\lambda = (1, 0.5)$. Posterior means on the natural scale, with 95% credible intervals in brackets.

Option	v_1	v_2	λ_1	λ_2	σ
A	1755.2 [1649, 1864]	-1341.6 [-1438, -1243]	0.926 [0.56, 1.37]	0.813 [0.50, 1.21]	104.85 [95.5, 115.4]
B	1951.8 [1885, 2024]	-1506.8 [-1565, -1444]	0.883 [0.55, 1.29]	0.770 [0.48, 1.12]	60.82 [53.3, 67.8]
C	1954.8 [1887, 2025]	-1511.2 [-1571, -1451]	0.886 [0.55, 1.30]	0.763 [0.47, 1.13]	59.61 [52.8, 67.1]
D	1952.6 [1881, 2024]	-1509.3 [-1572, -1448]	0.859 [0.53, 1.32]	0.782 [0.50, 1.10]	59.41 [52.8, 66.4]
truth	2000	-1500	1.00	0.50	50.0

The posterior mean for Option A places v_1 12.2% below the true value and σ 110% above it, with the true value outside the 95% credible interval in both cases. Options B, C and D place the velocities within 2.4% of the truth, with the true value inside the interval for both. Their posterior means for σ lie between 59.4 and 60.8, between 19% and 22% above the truth, with the true value below the lower interval bound in each case.

All four options place λ_2 between 0.76 and 0.81 against a true value of 0.50, with the true value at or just inside the lower interval bound. The estimated ratio λ_1/λ_2 lies between 1.09 and 1.16 for all four options, against a true ratio of 2.0.

Figures 5 and 6 present the same information as overlaid posterior densities and as intervals against the truth. The densities for Options B, C and D coincide closely for every parameter, differing by less than the width of the plotted lines for the velocities.

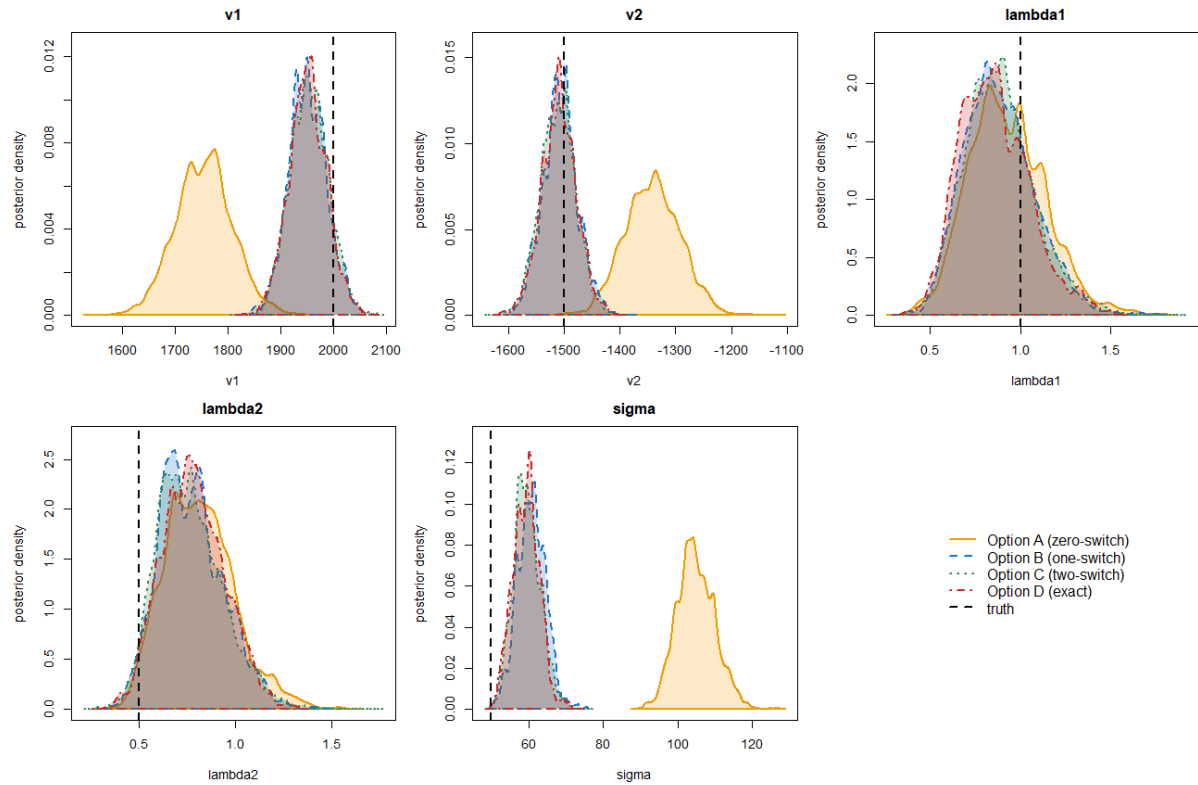


Figure 5. Posterior densities for each parameter, Options A to D overlaid. Dashed lines mark the true values.

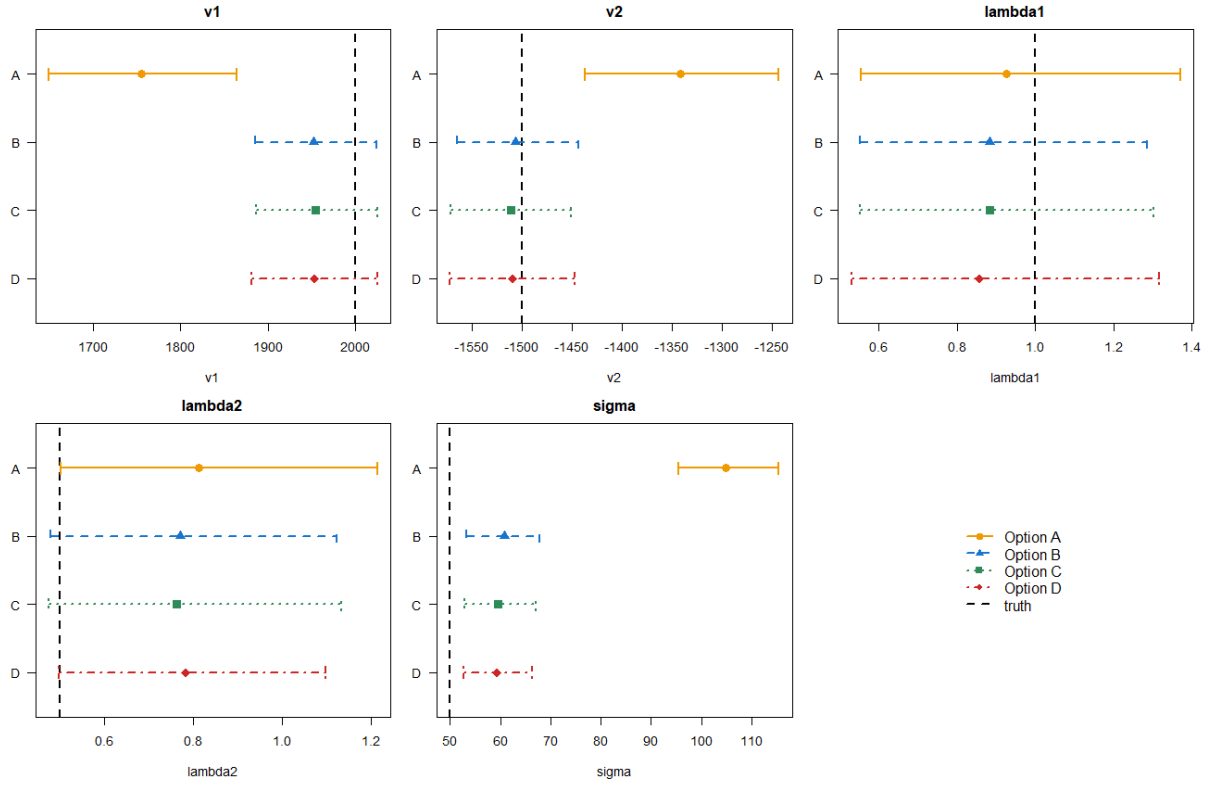


Figure 6. Posterior means and 95% credible intervals against the true values.

4.4 Faster switching

The baseline dataset contains few intervals with more than one switch, so it cannot distinguish the truncation orders. This experiment raises the switching rate to the point where it can.

Table 9. Faster-switching dataset, $N = 200$, $\lambda = (4, 2)$. 0.84 switches per interval, with 41 intervals containing two or more. The log-rate prior was widened to a standard deviation of 3 for this dataset, since the default prior is centred on a rate of 1.

Option	v_1	v_2	λ_1	λ_2	σ	$\max \hat{R}$
A	1301	-1100	2.60	1.53	136.3	1.007
B	1913	-1470	3.65	1.95	59.3	1.006
C	1949	-1494	3.64	2.17	52.6	1.015
D	1929	-1486	3.40	2.03	52.9	1.013
truth	2000	-1500	4.00	2.00	50.0	

All four options converged. Option A places v_1 35% below the true value and σ 173% above it. Option B places the velocities within 4.4% and σ at 59.3, 19% above the truth and essentially the same overestimate it produced at the baseline. Options C and D place σ at 52.6 and 52.9, 5% and 6% above the truth. Comparing C and D directly, C is closer to the truth on v_1 , v_2 and σ , D is closer on λ_2 , and B is closest of all on λ_1 . The estimated ratio λ_1/λ_2 lies between 1.67 and 1.87 for Options B, C and D, against a true ratio of 2.0.

4.5 Longer track

The preceding sections leave open whether the residual biases reflect the likelihoods or the quantity of data. This experiment lengthens the track eightfold at the same frame rate, so that the switching

rate and observation interval are unchanged and only the number of observations differs. All four options were run with six chains of 15,000 iterations, 5,000 discarded, and the velocity component of the proposal bound reduced to 5 for reasons given in Section 4.11.

Table 10. Longer-track recovery, $N = 1600$, $\lambda = (1, 0.5)$. The sampler for Option A located two modes, five chains settled in the principal one; its row is the mean over the five chains that agreed, following the convention stated in Section 2.10, with chain 5 reported separately below.

Option	v_1	v_2	λ_1	λ_2	σ
A	1754	-1396	0.94	0.48	93.8
B	2000	-1493	1.02	0.47	51.3
C	2021	-1498	0.99	0.49	50.4
D	2003	-1494	0.95	0.50	50.3
truth	2000	-1500	1.00	0.50	50.0

Options B, C and D place every parameter within 2% of its true value, including λ_2 between 0.47 and 0.50 and σ between 50.3 and 51.3. The five agreeing chains of Option A place v_1 12.3% below the true value and σ 87.6% above it, close to the corresponding figures at $N = 200$.

Table 11. Gelman-Rubin $\hat{R}$ by parameter at $N = 1600$. All four options were run under identical settings: 6 chains of 15,000 iterations, 5,000 discarded, and the default initial proposal covariance.

Option	v_1	v_2	$\log \lambda_1$	$\log \lambda_2$	$\log \sigma$
A	2.477	1.194	18.543	17.607	3.802
B	1.001	1.005	1.003	1.002	1.007
C	3.944	1.189	1.043	1.022	1.028
D	1.040	1.070	1.365	1.158	1.002

Option B's six chains agree on every parameter. For the remaining three the per-chain posterior means locate the disagreement. For Option A the sampler located two modes, five of six chains give v_1 between 1751 and 1756 with $\lambda_1 \approx 0.94$ and $\sigma \approx 93.8$, while chain 5 gives $v_1 = 1616$ with $\lambda_1 = 138.74$, $\lambda_2 = 28.19$ and $\sigma = 115.9$. For Options C and D all chains remain in a single mode, and the elevated statistic reflects slow mixing rather than separation, as for Option C, chain 2 gives $v_1 = 2108$ while agreeing on the other four parameters; for Option D, all six chains give $v_1 \approx 2003$ and $\sigma \approx 50.3$, with chain 2 giving $\lambda_1 = 0.81$ and $\lambda_2 = 0.55$ against 0.98 and 0.49 in the others.

4.6 The shape of the likelihood surface

The preceding sections report where the posterior places each parameter. This section examines the likelihood function itself, evaluated directly on grids of parameter values with the remaining parameters held at their true values, which shows whether the data constrain each parameter and where the likelihood is maximised.

One-dimensional slices, Option A

Figure 7 shows the Option A log-likelihood along each parameter axis in turn. The slices along v_1 and v_2 each have a clear maximum, but displaced from the true value in both cases, at approximately 1750 and -1250 against true values of 2000 and -1500. The slices along $\log \lambda_1$ and $\log \lambda_2$ have maxima close to the true values but are broad, changing by less than 25 log units across the plotted range, indicating that the rates are only weakly constrained by these data. The slice along $\log \sigma$ has no maximum within

the plotted range, as it rises steeply to approximately $\log \sigma = 4.5$ and is then flat to the edge of the range, so under this likelihood the data place little upper constraint on the noise scale.

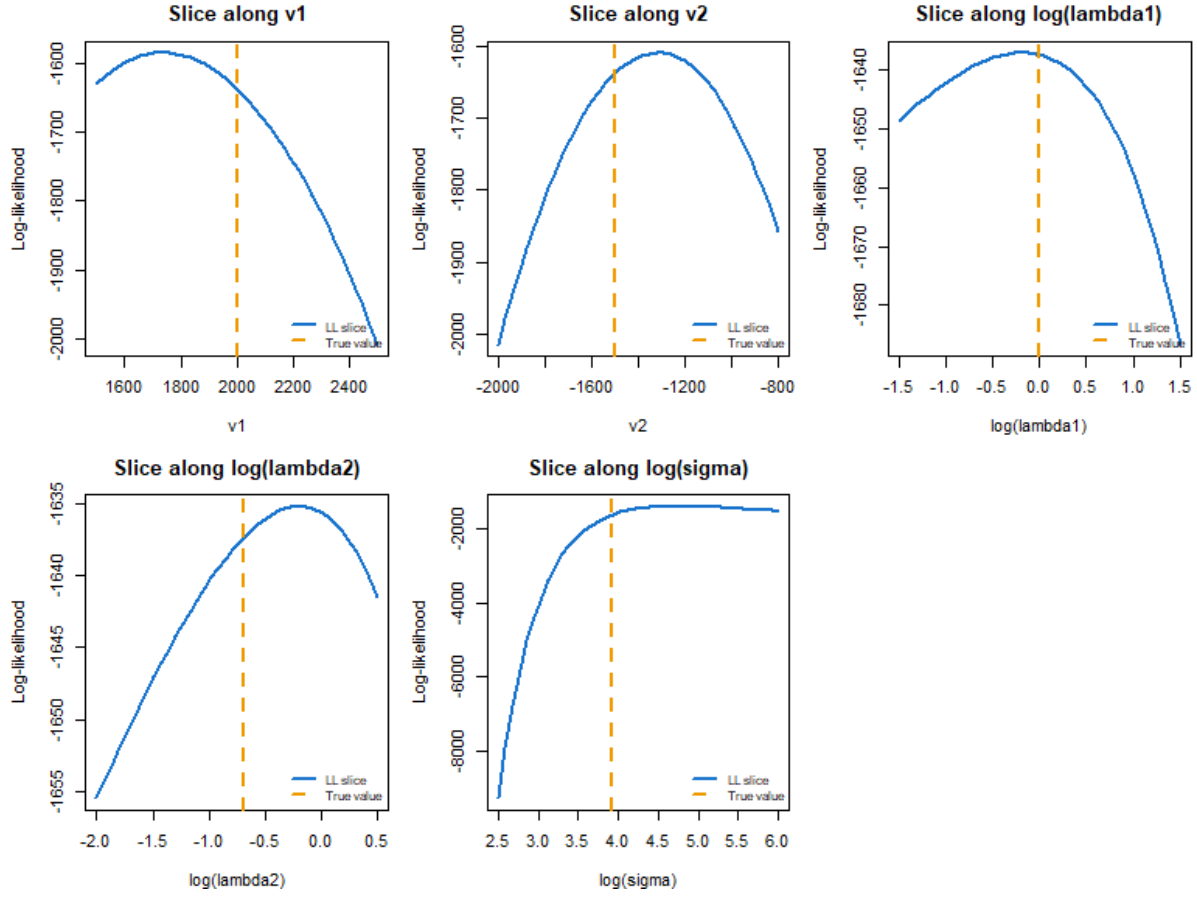


Figure 7. Option A log-likelihood along each parameter axis, other parameters held at their true values. Orange dashed lines mark the true values.

Joint surfaces over the velocity and the noise scale

Figures 8 and 9 show the joint log-likelihood over v_1 and $\log \sigma$ for Options A and D on the same baseline dataset, over the same parameter range.

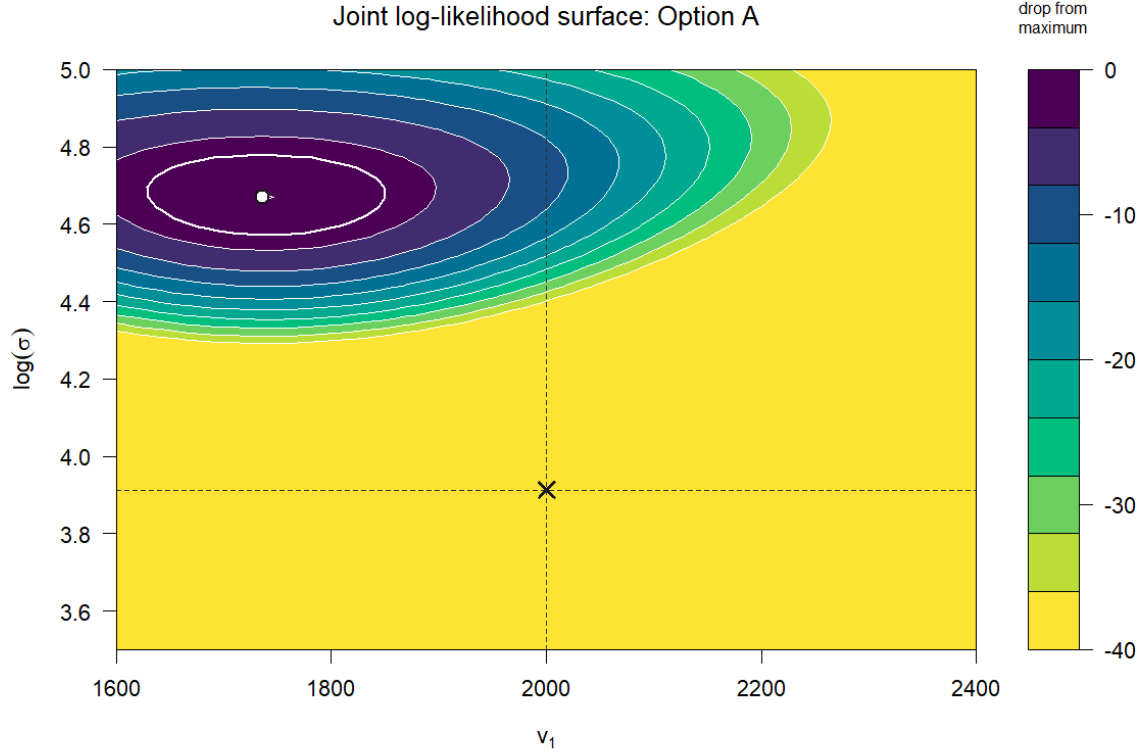


Figure 8. Joint log-likelihood over v_1 and $\log \sigma$, Option A, baseline dataset. Shading gives the drop from the maximum over the plotted grid, in log units, with values more than 40 below the maximum shown in a single colour. The white point marks the maximum, at $v_1 = 1736$ and $\sigma = 106.6$; the cross marks the true parameters. The true parameters lie far more than 40 log units below the maximum.

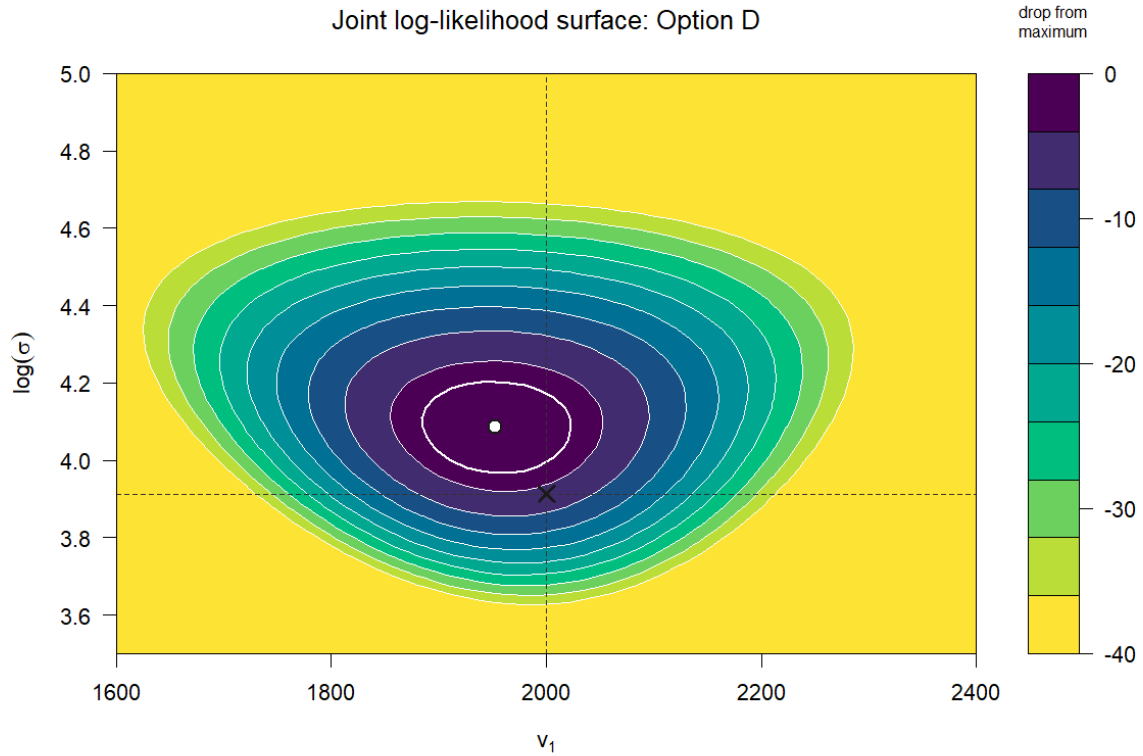


Figure 9. Joint log-likelihood over v_1 and $\log \sigma$, Option D, baseline dataset, on the same scale as Figure 8. The maximum is at $v_1 = 1953$ and $\sigma = 59.4$, and the true parameters lie 5.20 log units below it.

Both surfaces have a closed maximum within the plotted region, but in different places. For Option A the maximum lies at $v_1 = 1736$ and $\sigma = 106.6$, where the log-likelihood exceeds its value at the true parameters by 250.68 log units. For Option D the maximum lies at $v_1 = 1953$ and $\sigma = 59.4$, exceeding the value at the true parameters by 5.20 log units.

Table 12 reports a related comparison. Each likelihood was evaluated at the true parameters and at 20 parameter sets perturbed from them, drawn with standard deviations of 200 on the velocities and 0.3 on the log parameters.

Table 12. Log-likelihood at the true parameters and at 20 perturbed parameter sets, $N = 200$.

Option	At true θ	Mean perturbed	Best perturbed	Fraction below true θ
A	-1637.41	-1744.93	-1405.29	0.40
B	-1328.97	-1370.69	-1323.62	0.80
C	-1328.15	-1366.96	-1324.13	0.80

For Options B and C, 80% of the perturbed sets score below the true parameters, and the best exceeds them by 4.6 and 4.0 log units. For Option A, 40% score below the true parameters and the best exceeds them by 232 log units.

These figures are not directly comparable with the grid maxima above, since a random perturbation is unlikely to fall at the maximum; they bound the improvement from below rather than locating it.

The surface under fast switching

Figure 10 shows the Option D surface for the fast-switching dataset, $\lambda = (8, 4)$, over the same parameter range and with the same contour construction as Figure 9.

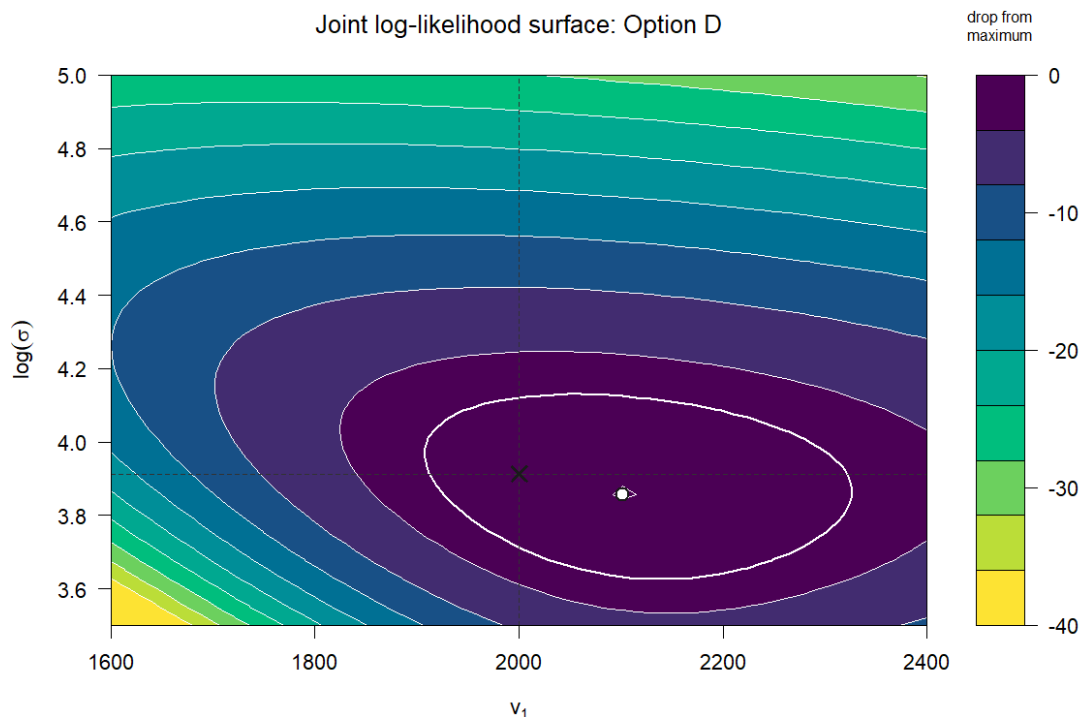


Figure 10. Joint log-likelihood over v_1 and $\log \sigma$, Option D, fast-switching dataset $\lambda = (8, 4)$, on the same scale as Figures 8 and 9. The maximum is at $v_1 = 2102$ and $\sigma = 47.3$, and the true parameters lie 0.56 log units below it. The region within a few log units of the maximum is far broader than in Figure 9.

The maximum lies at $v_1 = 2102$ and $\sigma = 47.3$, with log-likelihood -1406.00 against -1406.56 at the true parameters, a difference of 0.56 log units. The surface differs from Figure 9 in the range it spans, as across the same parameter box the log-likelihood varies by approximately 36 units here, against approximately 190 units at the baseline switching rate. The innermost contour encloses a correspondingly larger region of parameter space.

4.7 Holding the noise scale fixed

Since all four likelihoods overestimate σ at the baseline, this experiment tests whether that overestimate is what displaces the remaining parameters, by removing σ from the parameter vector and holding it at its true value.

Table 13. Fixed- σ experiment, N = 200, Option D.

σ treatment	v_1	v_2	λ_1	λ_2	σ	max $\hat{R}$
Estimated	1953	-1509	0.86	0.78	59.4	1.048
Held at truth	1962	-1518	0.94	0.81	50.0 (held)	1.032
truth	2000	-1500	1.00	0.50	50.0	—

Holding σ at the truth moves v_1 0.5% toward the true value, moves v_2 0.6% away from it, moves λ_1 from 0.86 to 0.94, and moves λ_2 from 0.78 to 0.81, further from the true value of 0.50. A separate attempt to constrain σ with a tightly concentrated prior rather than fixing it returned $\hat{R}$ between 1.35 and 3.88, so the chains did not settle in a common region, and the result is not used.

4.8 Sensitivity to starting values and priors

This experiment tests whether the rate estimate at the baseline depends on where the chains start or on the prior placed on the rates, for Option D. Each variation was run with six chains of 12,000 iterations, 4,000 discarded.

Table 14. Estimates of λ_2 under variations of the starting values and rate prior, N = 200, Option D. *The re-centred prior moves the prior median rate from 1 to 0.71, the geometric mean of the two true rates, so that the prior is centred closer to the truth.*

Variation	λ_2	max $\hat{R}$
Dispersed starting values (baseline)	0.77	1.005
All chains started at the true values	0.75	1.086
Rate prior widened, standard deviation 2 to 4	0.76	1.004
Rate prior re-centred on a median rate of 0.71	0.77	1.003

The estimate varies between 0.75 and 0.77 across the four variations, against a true value of 0.50. The re-centred prior returns the same value as the default, and its six chains agree to within 0.03 on every parameter.

4.9 Log-likelihood and computational cost

Table 15. Log-likelihood at the true parameters and cost per evaluation, N = 200. *Timings are the mean of 20 evaluations in one session.*

Option	Log-likelihood	ms per evaluation
A	-1637.41	3.70
B	-1328.97	19.56

Option	Log-likelihood	ms per evaluation
C	-1328.15	31.68
D	-1328.17	43.56

Options B, C and D agree to within 0.9 log units at the true parameters, and Option A is 308.4 log units below them. Across three sessions on the same machine Option D cost between 1.4 and 1.9 times Option C, while absolute timings varied by a factor of three between sessions. At $N = 1600$ the four options took 11.79, 72.45, 135.86 and 258.30 ms respectively.

4.10 Sensitivity to the integration grid

Table 16. Option D log-likelihood as the integration grid is refined, $N = 200$. The reference value is computed at 2049 grid points.

Grid points	Log-likelihood	Difference from reference	ms
65	-1328.6184	4.96×10^{-1}	23.06
129	-1328.3580	2.36×10^{-1}	12.91
257	-1328.2313	1.09×10^{-1}	22.68
513	-1328.1688	4.65×10^{-2}	36.82
1025	-1328.1377	1.55×10^{-2}	67.14

The log-likelihood changes by 0.11 units between 257 and 513 grid points, and by 0.03 units between 513 and 1025, the error falling by rather more than half at each doubling. The timings in the final column are not monotone in the grid size.

4.11 Proposal scaling

The lower bound on the proposal standard deviation, introduced in Section 2.9, proved consequential and its effect is recorded here because it determined which runs converged.

Table 17. Effect of the lower bound on the velocity component of the proposal standard deviation.

N	Posterior SD of v_i	Bound	Outcome
200	approx. 35	20	$\hat{R} = 1.16$ and 1.24 for the velocities
200	approx. 35	100	All $\hat{R}$ below 1.01
1600	approx. 15	100	Acceptance 0.001 to 0.03; $\hat{R}$ between 4 and 12
1600	approx. 15	5	$\hat{R}$ between 1.0 and 1.4

The bound that works at one sample size fails at the other. At $N = 200$ a bound of 20 was too small and the velocity chains drifted apart; raising it to 100 resolved this. At $N = 1600$ the posterior standard deviation of v_i is approximately 15, so a bound of 100 forced proposals far wider than the posterior and the acceptance rate collapsed; reducing it to 5 restored convergence.

Two further tuning changes were made during development. Adaptation was moved earlier and made more frequent, from beginning at iteration 1000 and updating every 200 to beginning at iteration 300 and updating every 100, which brought all baseline $\hat{R}$ values below 1.01. Supplying a hand-specified initial proposal covariance in place of the default was tried on two occasions and degraded mixing in both, as it returned $\hat{R} = 20.1$ for Option C at the baseline, and an earlier Option B run at $N = 1600$ using it reached $\hat{R} = 1.215$ against 1.007 for the same option under the default. The practice was abandoned.

4.12 Datasets with more than one posterior mode

On two datasets the sampler located more than one mode.

Fast switching, $\lambda = (8, 4)$. This dataset has 1.54 switches per interval, with 92 of 200 intervals containing two or more switches and 40 containing three or more. Options C and D were run with six chains of 15,000 iterations, 5,000 discarded. In both cases the sampler located two modes, five of six chains settled in a principal mode, and one chain in a second, giving maximum $\hat{R}$ of 2.227 and 24.549. These values are higher than those from an earlier run of four chains of 6,000 iterations, which gave 1.220 and 4.622, because the additional chains gave more opportunities to find the second mode rather than because the sampling deteriorated. The traces of the chains in the second mode oscillate about a stable level rather than trending, indicating that they have settled on a different solution and are mixing slowly within it. Options A and B are reported from the earlier run and were not repeated at the longer settings; in both, all chains settled in a single mode, returning $v_1 = 980$ with $\sigma = 140.7$, and $v_1 = 2083$ with $\sigma = 53.6$ respectively.

Table 18 gives the principal-mode estimates, computed as the mean of the individual chain means over the five chains that agree.

Table 18. Fast-switching dataset, $\lambda = (8, 4)$, principal-mode estimates, computed as the mean of the individual chain means over the five chains that settled there. *The fifth chain of each settled in a second mode and is reported separately below. These are not posterior means over all chains and are reported only because the fifth chain occupies a distinct region.*

Option	v_1	v_2	λ_1	λ_2	σ
C (5 chains)	2041.8	-1404.8	6.67	3.14	57.4
D (5 chains)	2124.2	-1425.0	7.54	3.34	53.9
truth	2000	-1500	8.00	4.00	50.0

The five agreeing chains of Option D are closer to the true values than those of Option C for v_2 , λ_1 , λ_2 and σ , with errors of 5.0%, 5.7%, 16.4% and 7.8% against 6.3%, 16.6%, 21.5% and 14.7%. Option C is closer for v_1 , with an error of 2.1% against 6.2%, and for the ratio λ_1/λ_2 , at 2.126 against 2.255 with a true ratio of 2.0.

The chains in the second mode differ from the principal mode substantially. For Option C, chain 5 returns $v_1 = 794$ and $\sigma = 160.1$. For Option D, chain 5 returns $v_1 = -454$, $v_2 = -3812$, $\lambda_1 = 83.1$, $\lambda_2 = 42147$ and $\sigma = 198.9$.

Another run was a thinned series, $k = 2$. Retaining every second observation gives $N = 100$ increments at $\Delta t = 0.6$. Option A's chains settled in a single mode, returning $\sigma = 227.5$ against a true value of 50. Options B and D each located a second mode, giving $\hat{R}$ of 3.994 and 1.938.

5 Discussion

5.1 Misspecification and its signature

At the baseline all four likelihoods return biased estimates of at least one parameter, and from Table 8 alone they appear to differ only in degree. The likelihood surfaces of Section 4.6 show that they differ in kind.

Option A is misspecified, as the emission it assumes is not the one that generated the data. Its emission assigns every interval a displacement of $v_s \Delta t$, so an interval containing a switch, whose true displacement lies somewhere between the two single-velocity values, can only be accommodated by inflating the estimated measurement error. The consequence is visible directly in the likelihood. Figure 8 shows a well-defined maximum at $v_t = 1736$ and $\sigma = 106.6$, where the log-likelihood exceeds its value at the true parameters by 250.68 units. Option A's likelihood is not vague about where the parameters lie; it is confident, and wrong. The perturbation test of Table 12 gives a weaker but independent version of the same result. It does not search systematically; it draws twenty random parameter sets and asks how they compare with the truth. For Option A the best of the twenty beats the true parameters by 232 log units, a lower figure than the grid maximum simply because a random draw is unlikely to land at the optimum, but the direction is unambiguous. For Options B and C, the best perturbation beats the truth by around 4 log units, which is the ordinary discrepancy between the true parameters and the maximum-likelihood point of a finite sample.

Figure 9 confirms that this is a property of the likelihood rather than of the data or the parameterisation. On the same dataset and over the same parameter range, Option D has a closed maximum at $v_t = 1953$ and $\sigma = 59.4$, only 5.20 log units above its value at the truth. Because the two figures share a colour scale, the contrast can be read directly from them, in Figure 8 the true parameters lie in the flat region beyond every contour band, more than 40 log units below the maximum, whereas in Figure 9 they sit inside the coloured region close to the maximum, a little over five log units below it. A further consistency check supports this reading. Option A's posterior is narrow and well defined, which is what a sharply peaked likelihood produces. Had its likelihood instead been flat over this range, the posterior would have reverted towards the prior, which is broad. The concentrated posterior and the sharp maximum of Figure 8 agree with one another, and both indicate a likelihood confidently maximised in the wrong place rather than one that is uninformative. The two surfaces differ because the emissions differ, and only one of them corresponds to the process that generated the data.

A distinction worth stating precisely is that three separate ideas are easily conflated here. Structural identifiability asks whether distinct parameter values give distinct data distributions, given unlimited perfect data; it is a property of the model. Practical identifiability asks whether the parameters can be recovered from a particular dataset, with its particular noise level, sampling interval and length. Misspecification is different from both, as it means the likelihood being fitted is not the law that generated the data. Option A's behaviour is misspecification. Its surface is not flat, and the problem is not that the model cannot in principle be identified; the problem is that Option A is confidently maximised in the wrong place, and no quantity of additional data will move it, as the unchanged estimates at $N = 1600$ confirm.

5.2 What the data can and cannot determine about the rates

Options B, C and D are correctly specified, yet at the baseline all three overestimate λ_2 by more than 50%. This bias behaves quite differently from Option A's, as it disappears at $N = 1600$, where all three

recover every parameter to within 2%. The explanation lies in the data rather than the method and has two parts.

The stationary distribution does not determine the rates

It is tempting to suppose that recovering the proportion of time spent in each state is nearly equivalent to recovering the rates. It is not. As noted in Section 2.1, the stationary distribution depends on the rates only through their ratio, so $(\lambda_1, \lambda_2) = (1, 0.5)$ and $(10, 5)$ are indistinguishable by occupancy alone. A particle may switch slowly and spend two thirds of its time in one state or switch ten times faster and spend the same two thirds there. Only the frequency of switching separates these, and that is what the observations must supply.

At the baseline the switching is barely observed. There are 50 interior switches in total, roughly 25 exits from each state, and 154 of the 200 intervals contain none at all. What suffers is precisely the ratio, as the estimated λ_1/λ_2 collapses toward 1, at 1.09 to 1.16 against a true ratio of 2.0, so the data can barely distinguish the faster-leaving state from the slower one. At $\lambda = (4, 2)$, with 168 switches, the estimated ratio recovers to between 1.67 and 1.87; at $N = 1600$, with 309 switches, the rates themselves are recovered.

The realised path is not typical of its own parameters

A second effect compounds this and is specific to the baseline realisation. As reported in Section 3.4, the hidden path spends 54% of its observation times in state 2, whereas the stationary distribution for its own generating rates is 67%. With only about 25 visits to each state, such a departure is unremarkable, but it has a measurable consequence.

If the hidden path were fully observed, the maximum likelihood estimate of each rate would be the number of exits from that state divided by the total time spent in it. Both quantities are recoverable exactly from the stored simulation output. The path spends 27.018 time units in state 1 and 32.982 in state 2, with exactly 25 completed exits from each, giving

$$\hat{\lambda}_1 = \frac{25}{27.018} = 0.925, \quad \hat{\lambda}_2 = \frac{25}{32.982} = 0.758$$

The estimates match the path that was simulated. These figures use only the known hidden path and no likelihood at all. The posterior means from Options B, C and D are 0.86 to 0.89 for λ_1 and 0.76 to 0.78 for λ_2 . The inference is recovering the rates of the realised trajectory to within about 4% for λ_1 and 3% for λ_2 ; what it is not recovering is the generating parameters, because this particular trajectory is not representative of them. The correct reading is that the dataset is too short for the realised path to reflect its own parameters, which is exactly what Ceccarelli et al. (2025) suggest when they observe that their best-fitting parameter set describes the data better than the true one does.

Section 4.8 rules out the alternative explanations. The estimate of λ_2 varies only between 0.75 and 0.77 whether the chains are dispersed or started at the truth, whether the rate prior is widened, or whether it is re-centred nearer the true values. Section 4.7 rules out one further candidate, as holding σ at its true value does not move λ_2 toward the truth but slightly away from it, so the noise overestimate and the rate bias are largely independent symptoms rather than cause and effect.

5.3 Practical identifiability and the limits of the data

Figure 10 shows what happens when the switching becomes fast relative to the frame rate. The Option D surface at $\lambda = (8, 4)$ is centred almost exactly on the truth, as its maximum lies 0.56 log units above the value at the true parameters, closer than at the baseline. The likelihood is not wrong. What has changed is that it is nearly flat. The contrast is visible in the extent of the near-optimal region. At the baseline the region within a few log units of the maximum is a compact ellipse occupying a small part of the plotted box. At $\lambda = (8, 4)$ the corresponding region extends across roughly half the range of v_i and runs off the lower edge of the plotted range in $\log \sigma$, so a wide range of parameter combinations describes the data almost equally well.

This is practical non-identifiability, as the model is being fitted correctly, but the data do not contain enough information to locate the parameters. This also explains the second mode reported in Section 4.12, though not in the way a flat surface alone would suggest. Running six chains of 15,000 iterations in place of four chains of 6,000 did not reduce $\hat{R}$ but increased it, from 1.220 to 2.227 for Option C and from 4.622 to 24.549 for Option D. The per-chain means show why, as in each case exactly one chain in six settles in a distinct region, so adding chains adds opportunities to find it, and lengthening them does not help a chain that has already settled there. The obstacle is therefore multimodality rather than slow exploration of a single region, and the flatness documented in Figure 10 is what makes the second region reachable rather than what prevents convergence directly.

The convergence of Options A and B on the same dataset should not be read as success. Option A returned $v_i = 980$ against a true 2000, converging confidently on a value that is plainly wrong, which is what Section 5.1 would predict. A likelihood that makes strong simplifying assumptions can present the sampler with a well-behaved posterior that happens to be in the wrong place, and this is easier to explore than a correct but flat one. Convergence and accuracy are separate questions.

Ceccarelli *et al.* (2026) examine identifiability for this class of model directly, and the behaviour observed here is of the kind their framework is designed to characterise.

5.4 The second mode and grounds for selecting between modes

Where the sampler locates a second mode, it takes a consistent form, in which a chain explains the data almost entirely as measurement error rather than as motion. At $\lambda = (8, 4)$, chain 5 of Option C settled at $v_i = 794$ with $\sigma = 160.1$, against $v_i \approx 2042$ and $\sigma \approx 57.4$ in the other five, so the forward velocity falls to about two fifths of its value elsewhere while the noise scale nearly triples. Chain 5 of Option D went further, reaching $v_i = -454$, $v_2 = -3812$, $\lambda_1 = 83.1$, $\lambda_2 = 42147$ and $\sigma = 198.9$. At $N = 1600$, chain 5 of Option A settled at $\lambda_i = 138.74$ with $\sigma = 115.9$, which is the same trade reached from the opposite direction, switching so fast that the two velocities average out, leaving a nearly constant effective velocity with the residual absorbed into the noise. In each case the sampler settles on the explanation that little structure is present and most of what is observed is noise.

The presence of this mode is not a failure of the analysis. It is a genuine feature of the posterior, and the same solution would exist for any method fitting this model, as an optimisation-based approach started from a similar point would report it as an optimum. The flatness documented in Figure 10

explains why it is reachable, since where the likelihood varies little across a wide parameter range a chain that drifts toward larger noise is not strongly penalised, and if it drifts far enough the velocities lose the contrast that would draw it back. Siekmann (2026, personal communication) reports the same behaviour in continuous-time Markov models of ion-channel gating, where the sampler assigns all observations to a single conductance level rather than resolving the open and closed states, so the phenomenon appears to be a feature of this class of hidden-state model rather than of the velocity-jump application.

What distinguishes the second mode from the principal one is that its parameters are poorly constrained, and this follows directly from its large noise estimate. Once the measurement error is permitted to absorb most of the variation in the data, the velocities and rates can take a wide range of values at little cost in likelihood. The oscillation visible in these chains reflects that lack of constraint rather than a failure to settle, as the chain is exploring a broad region, slowly.

There are therefore grounds for selecting the principal mode rather than just a preference for it. Both modes describe the observations, but the second does so only with wide uncertainty, whereas the principal mode constrains the parameters tightly and, on the simulated data used here, close to the values that generated them. If the purpose of the inference is to obtain parameters that can be used, the mode with the narrower posterior is the one to take. The second mode is reported rather than discarded, and it is best understood as a further instance of the practical non-identifiability discussed in Section 5.3, as it is what the posterior offers when the data do not constrain the noise scale from above.

Two points of restraint are appropriate. Whether this mode is peculiar to the exact likelihood was untested until the longer runs, which settle it, as Option C reaches the same region. The two differ in the severity of the excursion rather than in whether one occurs, Option D's chain in the second mode reaching a rate that removes the second state from the model altogether while Option C's remains within the plausible parameter range. And because the mode was observed only in regimes where Section 5.3 shows the data to be uninformative, its appearance may reflect the flatness of the posterior in those regimes rather than a general property of the likelihoods.

A separate and milder phenomenon appears at $N = 1600$, where chain 2 of Option D returns $\lambda_1 = 0.81$ against 0.98 elsewhere while agreeing on the velocities and the noise scale. This is not label switching, since the ordering constraint of Section 2.8 fixes the velocity labels and with the velocities ordered the rates are no longer exchangeable. It reflects the weak determination of the rates discussed in Section 5.2, appearing here as slow mixing rather than as bias.

5.5 The exact likelihood in relation to the approximations

Two datasets separate the truncations, and they separate different pairs. At $\lambda = (4, 2)$, where 41 of 200 intervals contain two or more switches, the one-switch truncation separates from the other two, as Option B leaves σ 19% high, essentially unchanged from its baseline error, while Options C and D reduce it to 5% and 6%. At $\lambda = (8, 4)$, where 92 intervals contain two or more switches and 40 contain three or more, the two-switch truncation in turn separates from the exact treatment. Each separation appears where the switch counts of Table 4 predict it should.

When the truncation order matters

At $\lambda = (4, 2)$, Options C and D are not distinguishable, as C is closer on three of the five parameters and D on one, with differences comparable to the Monte-Carlo error of the sampler. Nor should separation be expected there, since only 12 of 200 intervals contain three or more switches and there is little for the exact treatment to correct.

At $\lambda = (8, 4)$ the picture changes. Here 92 of 200 intervals contain two or more switches and 40 contain three or more, so the two-switch truncation is materially wrong for a substantial fraction of the data. In each case five of six chains agree, and comparing those consensus values shows Option D closer to the truth on four of the five parameters. The clearest separations are in the two quantities the truncation most affects, Option D recovers σ to within 7.8% against Option C's 14.7%, and λ_1 to within 5.7% against 16.6%. Option C remains closer on v_i and on the rate ratio.

This is the empirical separation that the lower switching rates do not provide. It must be qualified that the values compared are principal-mode estimates rather than posterior means over all chains, and the comparison rests on a single simulated dataset. It is suggestive rather than conclusive, and replication across independent datasets would be required to establish it firmly.

The case for the exact treatment

The case for Option D therefore rests on both structural and empirical grounds, though the empirical evidence is preliminary and is set out above. Structurally, it carries no truncation error by construction, so it requires no judgement about whether a given truncation order is adequate for a given switching rate, a judgement that in practice would require knowing the rates being estimated. Its cost is governed by the integration grid rather than by the switch count, so it does not grow as switching accelerates, whereas each additional order of truncation adds a further term. It costs roughly one and a half to twice the two-switch approximation, and Table 16 shows that coarsening the grid from 513 to 257 points costs 0.11 log units, negligible against the 308-unit gap between the switch-aware likelihoods and Option A.

On what is marginalised away

It might reasonably be asked whether Option D discards information by integrating over the occupation time, and in particular whether it loses track of which state each interval ends in, information the forward recursion uses. It does not. Because a two-state chain alternates, an odd number of switches ends in the opposite state and an even number ends in the starting state, so the two Bessel terms of Section 2.5 correspond exactly to the two possible end states. Keeping them separate yields the emission resolved by end state, and the recursion routes mass from start state to end state through it in the usual way. What is marginalised away is the switch count and the switch times, which carry no information about the parameters beyond what the occupation time already carries.

5.6 Proposal scaling

Table 17 records a result that is not specific to this model. A lower bound on the proposal scale exists to prevent an unrepresentative early covariance estimate from freezing the chain, but the bound that serves at one sample size fails at another, as 20 was too small at $N = 200$ and the velocities drifted, while 100 was too large at $N = 1600$, where the posterior is roughly half as wide, and almost every

proposal was rejected. The general lesson is that such a bound must be expressed relative to the expected posterior scale, which depends on the quantity of data, rather than fixed as a constant. The tilt of the contours in Figures 8 to 10, running diagonally rather than aligned with the parameter axes, indicates that the likelihood does not separate the velocity from the noise scale, as a change in one can be partly offset by a change in the other. A proposal that treated the parameters independently would therefore move across the contours rather than along them, which is why the velocity direction is the one that suffers when the proposal scale is poorly matched.

5.7 Comparison with Ceccarelli et al. (2025)

The datasets used here match theirs in the true parameters, the observation interval, the duration and the number of observations, so differences in results are attributable to the inferential machinery. Table 19 sets these out.

Table 19. Differences in inferential setup from Ceccarelli et al. (2025).

Aspect	This work	Ceccarelli et al. (2025)
Velocity prior	Normal, data-centred, SD 5000	Uniform [0, 4000] and [-3000, 0]
Rate prior	Normal on log scale, SD 2	Uniform [-4, 4] on log scale
Noise prior	Normal on log scale	Uniform [0, 100], natural scale
Adaptation	Rolling window during burn-in	Block-wise, recomputed per segment
Burn-in	2000 (4000 for Option C)	10000, then 10000 retained
Starting values	Dispersed about a data-derived centre	At the true values in several experiments

Two differences matter for the results. Their noise prior is uniform on [0, 100], so the Option A posterior mean of 104.85 obtained here lies outside their prior support, as the failure mode documented in Section 5.1 is one their configuration would have concealed rather than revealed, since the prior would have supplied the upper constraint that Option A's likelihood does not. Starting the chains from dispersed values rather than from the true parameters is the more consequential difference, since it is what makes the Gelman-Rubin diagnostic informative; chains all started at the truth may agree with one another without having explored the posterior.

Their reported rate regimes are consistent with what is observed here. They report valid estimates at $\lambda_1 = 2\lambda_2 = 1, 2$ and 4, a divergence between their two likelihood formulations at 8, and an inability to estimate the parameters accurately at 16. The multimodality encountered here at $\lambda = (8, 4)$ falls where they too report the problem becoming difficult, and Figure 10 supplies a concrete reason, that at that rate the likelihood surface is nearly flat.

5.8 Implications for the motivating application

Returning to the application of Section 1.1, the results carry a practical message about experimental design. What governs how well the switching rates can be recovered is not the number of frames as such but the number of switching events those frames capture. The baseline dataset contains 200 observations but only 50 switches, and the rates are poorly determined; the longer track contains 1600 observations and 309 switches, and they are well determined.

Two failure modes bracket the useful range. If the frame rate is too slow relative to the switching, most switches occur unobserved and the likelihood becomes flat, as Figure 10 shows. If the track is too short, too few switches are captured and the realised trajectory may not be representative of its own parameters, as Section 5.2 shows. For a given system, the interval should be short enough that intervals containing multiple switches are rare, and the track long enough to capture several hundred switching events. Where the frame rate cannot be increased, the exact likelihood at least removes the question of whether a truncation order is adequate, though it cannot recover information the sampling has discarded.

5.9 Limitations

- The derivation is specific to two states. For three or more the occupation-time law is multivariate and does not reduce to the two-state Bessel form, so the closed-form route taken here does not generalise directly.
- All results are from simulated data. The work establishes that the exact likelihood is correct and identifies where the approximations fail but does not demonstrate performance on experimental data.
- Each configuration was run on a single simulated dataset. Since Section 5.2 shows that a realised path may be unrepresentative of its parameters, replication across independent datasets would give a more reliable picture of each likelihood's bias.
- The comparison that most favours the exact treatment, at $\lambda = (8, 4)$, uses principal-mode estimates rather than posterior means over all chains, since in both options one chain of six settled in a second mode. The same applies to Option A at $N = 1600$. The grounds for selecting the principal mode are given in Section 5.4, but the affected results remain suggestive rather than established, and rest on a single simulated dataset.
- The timing comparisons are informal, taken from a small number of repetitions and varying by a factor of three between sessions.

6 Conclusions

This work derived an exact, truncation-free likelihood for the two-state stochastic velocity-jump model observed at discrete times with normal measurement error and compared it against the up-to- m -switch truncations at $m = 0, 1$ and 2 within a common Bayesian sampler.

The exact emission follows from the observation that the displacement over an observation interval depends on the hidden path only through the occupation time in the starting state, so that the sum over switch counts collapses to a single marginalisation over the occupation-time law, which for two states has a closed form in modified Bessel functions of the first kind. The parity of the switch count separates the two Bessel terms by end state, so the construction retains the start-to-end routing the forward recursion requires. It was confirmed against direct simulation, against exact normalisation, and against an independently implemented one-switch emission to which it converges as the integration grid is refined.

Three findings emerge:

- The zero-switch approximation is misspecified, not just imprecise. Its likelihood is maximised at $v_1 = 1736$ and $\sigma = 106.6$, 250.68 log units above its value at the parameters that generated the data, and its estimates are essentially unchanged when the track is lengthened eightfold. The exact likelihood on the same data is maximised 5.20 log units above the truth.
- The one-switch truncation is adequate at low switching rates and insufficient at higher ones, and the two-switch truncation in turn becomes insufficient at higher rates. The one-switch treatment leaves the noise scale 19% high at $\lambda = (4, 2)$, where the two-switch and exact treatments reach 5% and 6%. At $\lambda = (8, 4)$, where 92 of 200 intervals contain two or more switches, the exact treatment recovers the noise scale to within 7.8% and the faster rate to within 5.7%, against 14.7% and 16.6% for the two-switch truncation, though this comparison uses principal-mode estimates, five of six chains having settled there in both options.
- The residual bias in λ_2 at $N = 200$ comes from the data, not the method. It survives every starting-value and prior variation tested, it matches the estimates computed directly from the realised hidden path, and it disappears when the track is lengthened.

A fourth observation concerns the limits of the approach. When switching becomes fast relative to the frame rate, the exact likelihood remains centred on the true parameters, its maximum lying 0.56 log units above the value at the truth, but the surface becomes nearly flat, varying by about 36 log units across a parameter range over which it varies by 190 at the baseline. The parameters are then not practically identifiable, and the appearance of a second mode reflects that rather than any defect in the likelihood. Increasing the number of chains and quadrupling their length increased rather than decreased $\hat{R}$, because the difficulty is a second mode that additional chains are more likely to find, not insufficient exploration of a single one. An inflated $\hat{R}$ in this situation reflects the separation between modes rather than a failure to equilibrate, and the modes are therefore reported separately. This is the more useful framing of the limit, as the exact likelihood does not fail in this regime, but neither can it extract information the sampling has not captured.

At the rates where a single mode was found, Option D produced no more accurate estimates than the two-switch truncation, and at those rates the evidence does not support a claim of superior accuracy. The one comparison that favours it, at $\lambda = (8, 4)$, uses principal-mode estimates rather than posterior means over all chains, so it is suggestive rather than established. The advantages that do not depend on it are that Option D requires no judgement about the adequacy of a truncation order, a judgement that would depend on the very rates being estimated, and that its cost is governed by a tuneable integration grid rather than by the switching rate.

A broader methodological point follows. Parameter bias observed at a single sample size cannot by itself distinguish a misspecified model from a correct model applied to limited data, yet the two demand opposite responses. Separating them here required examining the likelihood surface directly and lengthening the track, and neither alone would have sufficed.

Future work

- Replicate each configuration across independent simulated datasets, to separate each likelihood's bias from the peculiarities of a single realised path.
- Replicate the $\lambda = (8, 4)$ comparison across independent simulated datasets and investigate whether a sampler less prone to the secondary mode, such as one using tempering or multiple restarts with mode filtering, allows the comparison to be made without recourse to mode selection.
- Map the likelihood surface across a range of switching rates, to locate the point at which practical identifiability is lost.
- Adopt a proposal respecting the correlation between the velocities and the noise scale, or a gradient-based sampler, to address the geometry visible in Figures 8 to 10.
- Generalise the occupation-time derivation beyond two states.

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Self-evaluation

Approach

Reproducing an existing method before attempting to extend it was the most useful decision of the project. Implementing the truncated hierarchy first meant that when the exact expression was derived there was already an independently written reference to test it against, and the strongest of the confirmation checks, the convergence of Option D's one-switch truncation onto the Option B emission, exists only because that work was done in that order.

Confirming the derivation against quantities computed by routes that do not use it was equally important. Direct simulation of the occupation time and exact normalisation are two independent routes, and the internal requirement that the two Bessel terms sum to the total density guards against the specific error that did occur. Agreement across all three makes the derivation defensible in a way that a single check would not.

Course corrections

Several results had to be revised as the work progressed, which is a normal feature of a project of this kind rather than a set of avoidable errors, but the sequence is worth recording.

The increment variance. An early version of the pipeline used σ^2 rather than $2\sigma^2$ as the variance of an observed increment, overlooking that differencing two independently noisy positions adds their variances. The symptom was a noise estimate near 148 against a true value of 50, and the diagnosis followed from noticing that 148 is close to $\sqrt{2} \times 104.85$, the estimate obtained after the correction, which is exactly the factor the error implies. The derivation now appears in full in Section 2.2.

Sampler tuning. Time was spent on proposal-scale problems that were initially read as problems with the model. A bound of 20 allowed the velocities to drift; raising it to 100 fixed that but exposed poor rate mixing; a global scale multiplier of 0.6 made matters worse and was reverted; moving adaptation earlier resolved it. At the longer track length, the same bound of 100 was then far too wide. The general principle, that such a bound must scale with the quantity of data, only became visible once the failure had been seen in both directions, and it is reported in Section 4.11 as a result rather than as an implementation note.

A cost claim. An early benchmark showed Options C and D costing almost the same per evaluation, which was written up as evidence that the truncation hierarchy crosses over at $m = 2$. Repeated benchmarking did not support it, as the ratio is between 1.4 and 1.9, and the original figure came from a single noisy measurement. The claim was withdrawn and replaced with the structural argument the evidence does support.

Judgement about what could be claimed

The recurring difficulty was separating what the runs supported from what would have made a tidier narrative. The comparison was initially framed as a ladder of increasing accuracy ending in Option D, which the faster-switching results do not support, since C and D are indistinguishable there; the framing was replaced and the case for D restated in structural terms. Option C's baseline estimates were nearly quoted despite an $\hat{R}$ of 1.115 and were withheld until a longer run brought the diagnostic below threshold. The hypothesis that the inflated noise estimate was distorting the other parameters was intuitive, was tested directly, and was not supported; it is reported as a negative result.

The most substantial revision concerned the fast-switching dataset. The elevated convergence statistic for Options C and D there was at first read as a limitation of those likelihoods. Plotting the likelihood surface showed that the surface is correctly centred but nearly flat, which reframed the finding entirely, that the limitation belongs to the data, not the method. This was also the point at which the distinction between misspecification and practical identifiability became clear, and an earlier draft that described Option A's behaviour as an identifiability problem was corrected. A related correction followed later. The runs at $\lambda = (8, 4)$ and the largest-sample runs were described as having failed to converge, based on an elevated Gelman-Rubin statistic. The statistic assumes that the chains are converging to a common distribution, so where several chains settle in one mode and one settles in another it reports the separation between them rather than a failure to equilibrate. The traces of the chains concerned oscillate about a stable level rather than trending, indicating that they have settled. The results are now presented as the sampler having located more than one mode, with the modes reported separately, which is both a more accurate description and a stronger basis for the comparison in Section 5.5.

What I would do differently

- Plot the likelihood surface early. It answered in one figure a question that several MCMC experiments had left open, and at a fraction of the computational cost.
- Establish conventions for reporting convergence and for recording run settings before running experiments rather than after. The convention adopted here was revised twice, and both revisions would have been avoided by thinking through what the diagnostic does and does not measure at the outset.
- Simulate several independent datasets per configuration from the start. Much of the interpretive difficulty came from reasoning about a method's behaviour from a single realisation of a stochastic process.
- Run the longer-track experiment sooner. It settled a question that consumed considerable effort at the baseline, and the diagnostics undertaken to eliminate competing explanations would have been better motivated by a result already in hand.
- Read the identifiability literature earlier, since it provides the vocabulary that several of these observations required.

Generative AI statement

I used generative AI tools (Anthropic's Claude) during this project.

Code implementation

The approach taken in Module 2D, namely that the displacement over an observation interval depends on the hidden path only through the occupation time in the starting state, and that the sum over switch counts can therefore be replaced by a single marginalisation over that occupation time, is my own. The closed-form density of the occupation time for a two-state chain is a standard result (Pedler, 1971); I used AI assistance to locate it and to confirm that its series expansion reproduces the switch-count sum term by term. AI was used to implement, optimise and debug that derivation and the surrounding pipeline in R, as well as to draft the explanatory comments.

Numerical and computational techniques

I used AI assistance to identify and implement several techniques beyond my prior experience in R:

- The vectorised convolution used in Modules 2B, 2C and 2D, in which the integral of the emission density against the measurement-noise density is expressed as an outer-difference matrix multiplied by a vector of quadrature coefficients, `dnorm(outer(delta_y, dx_grid, "-")) %*% coef`, rather than looping over increments. This gave a substantial reduction in runtime with an identical result.
- Cholesky factorisation of the proposal covariance in Module 3, so that multivariate normal proposals are drawn as `theta + crossprod(chol(Sigma), z)` with `z` standard normal, and the factorisation is computed once outside the sampling loop rather than at each iteration.
- Construction of the three-dimensional emission tensor $E[i, s1, s2]$ in Modules 2B, 2C and 2D, and the partial array indexing used to accumulate the zero-switch atom and the two parity components into the appropriate slices.
- Construction of the composite Simpson quadrature weights by strided vector assignment rather than by iteration.
- Computation of the matrix exponential by eigendecomposition of the generator, together with the subsequent handling of negligible imaginary parts, clipping of small negative entries and row renormalisation.
- Solution of the stationary distribution, by eigendecomposition of the transposed generator in Module 2B and by `qr.solve` on the augmented balance equations in Module 2D, with a fallback to a uniform distribution if the system is numerically singular.
- The adaptive Metropolis scheme of Module 3, specifically the rolling-window covariance estimate, the $2.38^2/d$ optimal scaling factor of Roberts and Rosenthal (2001), the small ridge term maintaining positive definiteness, the lower bound applied to the covariance diagonal, and the recovery step that retains the previous covariance if a Cholesky factorisation fails.
- Implementation of the Gelman-Rubin statistic directly from its definition rather than through an existing package.
- Assignment of exact switch times to observation intervals using `findInterval` and `tabulate`.
- The compact test for the velocity ordering constraint, `is.unsorted(rev(v), strictly = TRUE)`.
- The closed-form normalising constants for the one-switch and two-switch densities in Modules 2B and 2C, including the separate branches required when the two switching rates are equal.
- The choice of integration domain and quadrature grid in Module 2B, restricting the convolution to the compact support of the exact-increment density, on the reasoning that the measurement-noise tail outside the support is captured by evaluating the noise density

at grid points within it, and that integrating across the discontinuity at the edges of the support would lose accuracy.

- The use of tolerance-based rather than exact comparisons in the degenerate-case guards in Modules 2B, 2C and 2D, for equal velocities and equal switching rates, since parameter values proposed by the sampler are never exactly equal in floating point.
- The guards controlling which code executes when a module is sourced by another, using `sys.nframe() == 0` and `exists("sim")`.

Errors identified and corrected with AI assistance

Three substantive errors were found this way:

- In the first implementation of Module 2D, the two Bessel terms of the occupation-time density were transposed, so that the mass ending in the start state and the mass ending in the other state were assigned to the wrong end states. The error was identified by comparing the implementation against an end-state-resolved Monte-Carlo simulation of the underlying chain and corrected. The parity decomposition as now reported, in which the I_0 term carries the odd switch counts and the I_1 term the even, is the corrected form.
- The lower bound on the velocity component of the proposal standard deviation was found to fail in both directions. At $N = 200$ a bound of 20 was too small and the velocity chains drifted apart, giving $\hat{R}$ values of 1.16 and 1.24. At $N = 1600$ the bound of 100 that had resolved this was approximately seven times the posterior standard deviation of v_i , and the acceptance rate collapsed to between 0.001 and 0.03. Diagnosing both failures and arriving at the general principle that such a bound must scale with the quantity of data, was done with AI assistance and is reported in Section 4.11.
- In Module 4, the chain colour palette was indexed such that chains 5 and 6 of a six-chain run were drawn in a fully transparent colour and were therefore invisible in the trace plots. This was identified and corrected.

Figure presentation

I used AI assistance in selecting a colour-blind-safe palette for the figures, and in pairing colour with distinct line types so that the figures remain interpretable when printed in black and white.

Preparation of the report

AI was used to convert the mathematical expressions in the report into LaTeX for insertion into Microsoft Word's equation editor, and to compile the digital object identifiers for the reference list, which I then verified individually against the publisher records. No report text was written by the AI.

Verification

All AI-assisted code was verified by execution and by comparison against independent references that do not reuse the derivation being tested. These include direct Monte-Carlo simulation of the occupation time and end state from 400,000 simulated paths, exact normalisation of the emission density, the internal requirement that the two parity components sum to the total density, and agreement between the one-switch truncation of Module 2D and the separately implemented one-switch density of Module 2B as the integration grid is refined. The results of these checks are reported in Section 4.1.

Appendix

Post-burn-in trace plots for the three baseline runs not shown in Section 4.2, all from the $N = 200$ baseline dataset. Options A and D were run with four chains of 6,000 iterations, giving 4,000 post-burn-in samples per chain. Option C is the longer baseline run described in Section 4.2, with six chains

of 12,000 iterations and 4,000 discarded, so its post-burn-in series is 8,000 iterations; it is not one of the $N = 1600$ runs of Section 4.5.

Trace plots — Option A (4 chains, post burn-in)

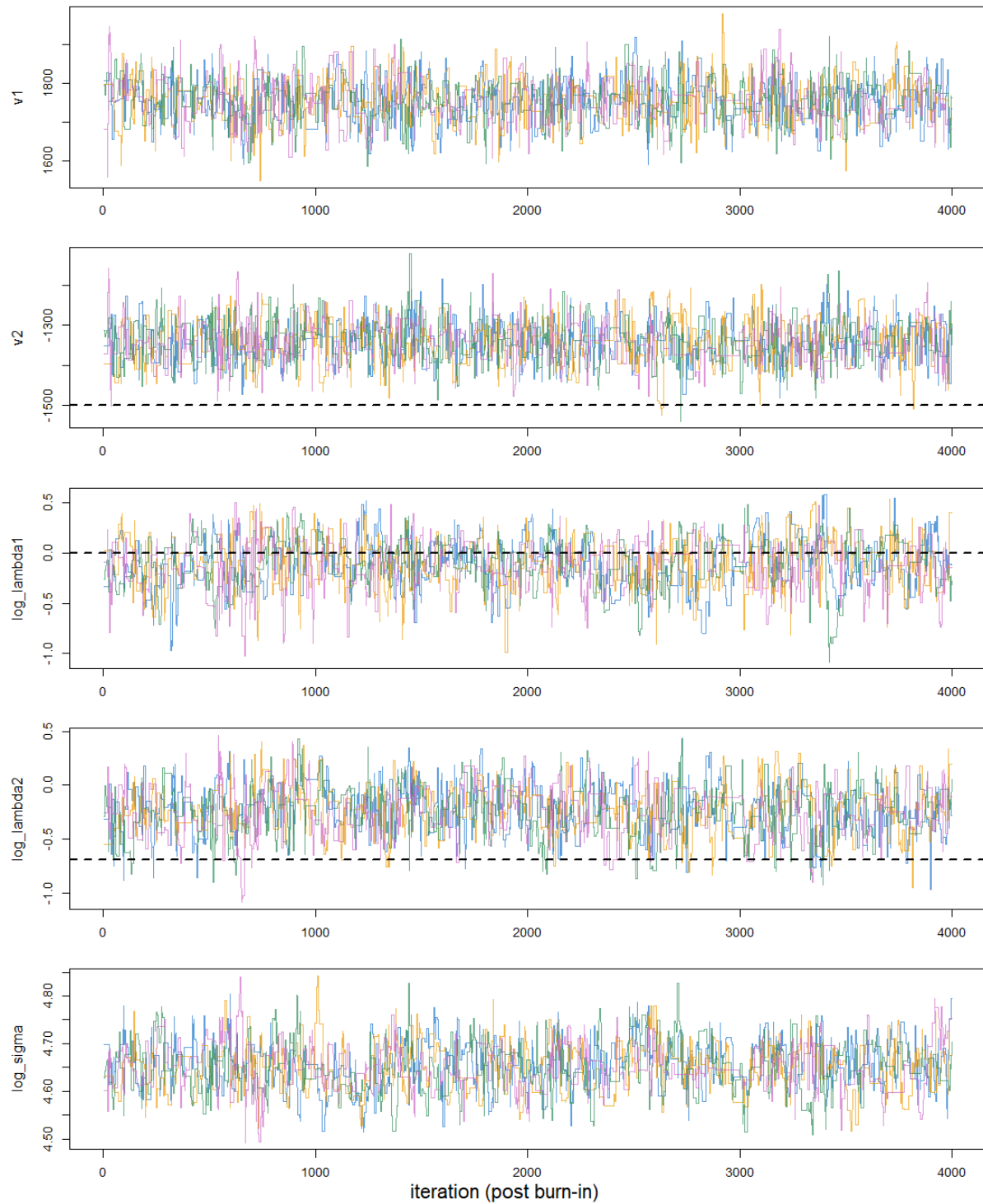


Figure A1. Trace plots for Option A, four chains, post burn-in. Dashed lines mark the true values.

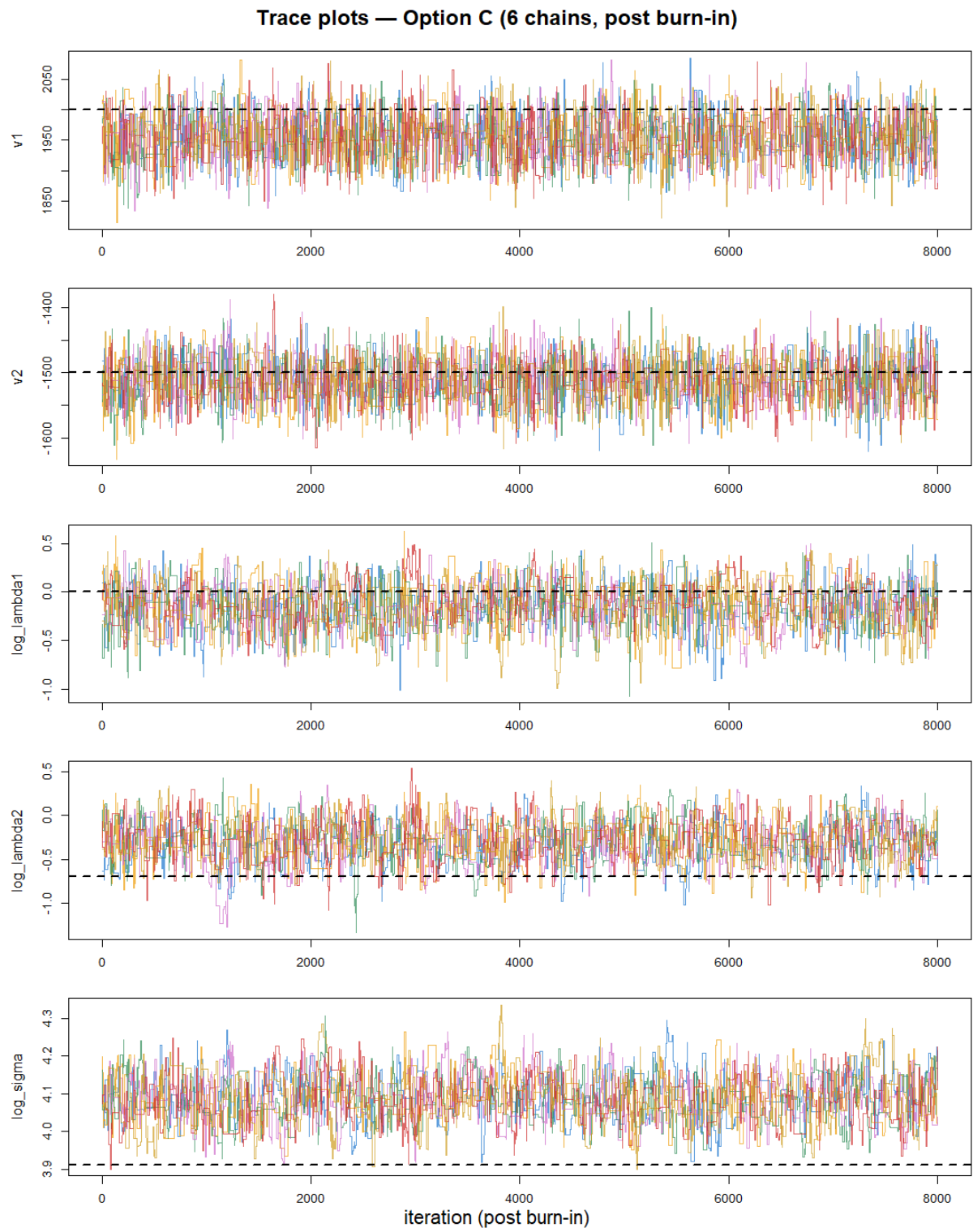


Figure A2. Trace plots for Option C, six chains, post burn-in.

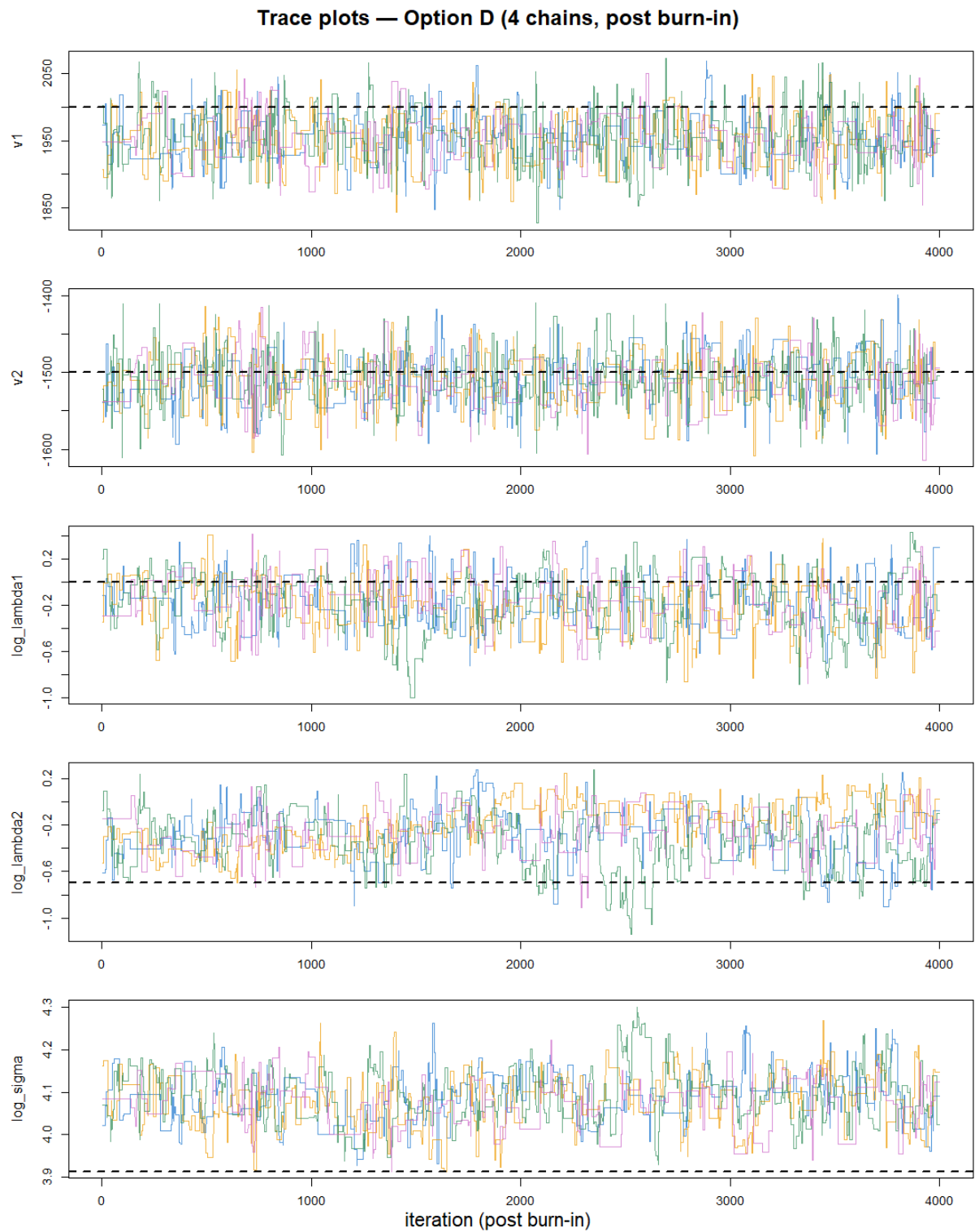


Figure A3. Trace plots for Option D, four chains, post burn-in.